A DATA-DRIVEN INVESTIGATION

THE DISCARD

Why America throws away the kidneys it begs for, and the inventions waiting to stop it



EVERY 100 KIDNEYS DONATED IN THE UNITED STATES, 2024



29 WENT UNUSED

Source: OPTN/SRTR 2024 annual data report

JEFF PARKE

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COLD ISCHEMIA FOUNDATION

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For the families who said yes in the worst hour of their lives, and for the people still waiting for that yes to arrive.

"He that will not apply new remedies must expect new evils: for time is the greatest innovator."

— Francis Bacon, "Of Innovations" (1625)

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Prologue: Offer Fourteen

> *"Life is short, and the Art long; the occasion fleeting; experience fallacious, and judgment difficult."*
>
> Hippocrates, Aphorisms, section 1 (Adams translation, 1849)

The phone rings at 11:42 on a Tuesday night, and it is not Denise's phone.

It rings in a transplant center forty minutes from her house, on the desk of a coordinator who has done this for eleven years and can tell from the first syllable what kind of call it is. A donor has died in a hospital two states away. The family said yes. The kidneys have been recovered, packed in ice, and offered, through a national computer system that sorted every person waiting in the country by blood type, tissue match, years on the list and a dozen smaller things, to the people at the top of the list. Denise is near the top. She has been on dialysis for six years and on the list for five. She is fifty-four. She teaches fourth grade, and she has not missed a Monday, a Wednesday or a Friday treatment in all that time, because she has been told, over and over, that being reliable is how you get chosen.

She is asleep. Her phone is on the nightstand, face down.

Go back eight hours, to the other end of the night. In a hospital two states away a woman in her sixties is sitting beside a bed, holding a hand that is warm and will stay warm for a while yet. A neurologist has explained, twice, in plain words, what the tests show. A different person, a stranger with a lanyard and a soft voice, has asked whether the family has ever talked about donation. The woman says that her husband carried the little red heart on his driver's license for thirty years. She says yes. She signs. She will go home before the kidneys leave

the building, and she will not know, for a very long time, what became of them. Perhaps she never will. The families who say yes are often told very little about what happened to the gift.

Surgeons arrive near midnight. The kidneys come out in the practiced, quiet way that strangers have been learning for seventy years, are flushed with cold fluid, and sealed in sterile bags inside a box of ice. From that moment a clock starts that no one in the building can see. The kidney has no blood flowing through it. It is living on what it stored, and on cold, and every hour that passes is an hour it will carry into whoever receives it.

The coordinator reads the screen aloud to the surgeon on call. The donor is fifty-eight. He had high blood pressure. The kidney was biopsied at the donor hospital, and a pathologist, reading a frozen slice of tissue in the middle of the night, saw some scarring. The kidney will have been without blood flow for about nineteen hours by the time it could reach the operating table. It is a kidney with a few warning signs and a good chance anyway, the kind that clinicians argue about, the kind that on any given night one team will take and another will not.

The surgeon is tired. She was in the operating room until nine. She thinks about Denise, who has diabetes, who is not young, who will be a little sicker than average on the table. She thinks about the number the government publishes every six months about how many of her patients' new kidneys are still working a year later, and about what happens to a program when that number slips below what a formula predicted. She thinks, because she is honest, that she may be right to say no. The kidney might not work right away. Denise might spend two weeks in the hospital she is trying so hard to leave.

She says: *Decline.*

The coordinator types a reason into a drop-down menu. It takes eleven seconds. Somewhere in a data center a list re-sorts itself, and the kidney is offered to the next person, and the next. By morning it will have been offered to a dozen people, in a dozen centers, with a dozen versions of the same thought.

At 7:10 Denise wakes up, checks her phone, finds nothing, and drives to her chair.

I have never met Denise. She is invented, and I am telling you so now because the rest of this book will insist on the difference between what is invented and what is counted. The night is a composite. Its parts are not. In a study of about fourteen million kidney offers made between 2008 and 2015, researchers found that candidates who died without a transplant had received a median of sixteen offers over 651 days on the list (Source Notes S1). Denise's call was offer fourteen. In the great majority of declines, the centers recorded the reason as something wrong with the organ or the donor. Some of those declines were surely right. The researchers said so, and so will I.

But sit with the arithmetic for a moment. In 2024, according to the national registry, 29.3 percent of the kidneys recovered from deceased donors in the United States were never transplanted into anyone (Source Notes S2). Eleven years earlier the figure was 18.2 percent. The families said yes. Surgeons drove through the night. Planes flew. And nearly three in every ten of those kidneys never reached a body.

That is the number this book follows.

I did not set out to write an angry book, and I do not think this is one. Everyone in the story I am about to tell is, as far as I can find, trying to do the right thing: the surgeon who says no, the coordinator who types the reason, the regulator who writes the rule, the family who signed the form. The tragedy, if there is one, is that a system made of decent people can arrange itself so that the safest choice for each of them is the costliest choice for someone they will never meet.

It is also not a story without hope. Along the edges of the problem sit tools that were proven years ago and never fully used: a pump that keeps a kidney alive in transit, and that, in a ten-year follow-up of a randomized trial, left more kidneys working a decade later. A molecule borrowed from a marine worm that carries forty times more oxygen than our own, by its makers' account. A fleet of new dialysis machines that, in a large European trial, lowered the death rate, and that American patients are only now beginning to see. A memo, sent on a single day in August of 2025, that changed the behavior of an entire national network within weeks, and in the wrong direction.

You will meet all of them. By the end I want you to be able to answer three questions without notes: where the kidneys go, why they go there, and what, specifically, would stop it.

Denise, in the meantime, has an appointment with her nephrologist next week. She will ask whether anything has come up. She will be told, kindly, that she is still on the list.

Introduction: The Invisible Ledger

> *"The more any quantitative social indicator is used for social decision-making, the more subject it will be to corruption pressures and the more apt it will be to distort and corrupt the social processes it is intended to monitor."*
>
> Donald T. Campbell, "Assessing the Impact of Planned Social Change" (1979)

A kidney is a gift with a deadline.

It comes from a person who has died, usually without warning, and from a family that agreed in the first hour of its grief to let a stranger keep living. It leaves one body and has to reach another within a window the surgeons measure in hours. Every hour it spends in the cold, without blood flowing through it, adds a little damage that the recipient will carry. That damage is real, measurable, and in many cases avoidable. It is also not the only way a kidney is lost. It can be lost in a cooler. It can be lost in a queue. It can be lost in a phone call that takes eleven seconds.

This book is about that loss. It makes a claim that I intend to defend with every source I can find, and I would rather state the claim now than make you hunt for it.

The American transplant system throws away roughly three in every ten of the kidneys it recovers, not mainly because the kidneys are bad, but because the system rewards the people who say no and punishes the people who say yes; and the tools that could change the arithmetic have been proven, published, and left on the shelf.

Each half of that sentence deserves a warning label.

The first half does not say that every discarded kidney was good. Many were not. Donors are older and sicker than they used to be, as the country is, and a surgeon who declines a badly scarred kidney is doing her job. The claim is narrower and harder to dismiss. It is that the *rate* of nonuse has climbed from 18.2 percent in 2013 to 29.3 percent in 2024 (Source Notes S2), that a French analysis of American discards found that most of them, by the standards of French practice, would have been transplanted (Source Notes S3), and that the people who decline kidneys are operating inside a structure whose incentives lean one way. I will show you that structure piece by piece.

The second half does not say that every new technology works. Some do not. In 2022 the largest trial ever done in dialysis tested a cheap, appealing idea, cooler dialysis fluid, in more than fifteen thousand patients, and found it did nothing for the heart (Source Notes S4). I will tell you about that trial, because a book that tells you only the stories that flatter its argument is not worth reading. What I am claiming is that for a short list of tools, the evidence has been in hand for years, and that the delay is better explained by who pays, who is blamed, and who owns the problem than by any doubt about the science.

Two ledgers

The idea that organizes this book is simple enough to say in a minute and to fight about for a year.

Imagine every decision in the transplant system recorded in one of two ledgers. In the first, the visible ledger, go the things that can be counted and blamed. A surgeon accepts a marginal kidney; it fails in eight months; the failure is recorded, attributed to her program, and published. A procurement organization recovers organs from a donor whose neurological status turns out to be uncertain; a federal investigation follows. A dialysis center spends more on a patient than the formula allows; the margin shrinks. All of these are entries in the visible ledger, and each has a name, a date, and an owner who can be held to account.

In the second, the invisible ledger, go the things that cannot. A kidney declined for the fourteenth time goes unused. A candidate who would have lived five more years with it dies on the list in the spring. A pump that would have kept a kidney healthy is left in a warehouse because no one in the chain gets credit for using it. There is no row for these, no name, no owner. The loss is real, but it is distributed so thinly across so many hands, and spread so far in time, that it never lands on anyone's desk.

The Campbell epigraph above says what happens next, and it says it more bluntly than I could. When you measure a social process and attach consequences to the measure, people respond to the measure. The measure here is the program's one-year outcome, counted in the visible ledger. The thing we actually care about, how many people end up with a working kidney, lives in the invisible one.

I will argue that almost everything strange about the American kidney system, including the discards, the delay in adopting proven tools, and the oddly deliberate pace at which good ideas arrive, can be read as a consequence of one asymmetry. *The system is built to notice a yes that goes wrong, and to ignore a no that goes wrong.*

Four ideas to carry through

I will use four names, and I will use them often.

The invisible ledger is the asymmetry just described: the costs of acting are recorded and attributed; the costs of not acting are neither.

The ratchet is how the invisible ledger gets stronger over time. Every scandal produces a rule or a metric or a memo meant to prevent the last mistake, and almost none is ever withdrawn. Caution accumulates. A system that adds a guard against every visible failure and never subtracts one will, given enough years, guard itself into paralysis. We will see a ratchet turn in chapter 5, in a span of forty-eight days.

The orphan innovation is a tool that works but has no parent: no one whose budget it fits, no one whose metric it improves, no one who is paid for the hours it saves. A machine that benefits the recipient but is paid for out of the procurement organization's budget is an orphan. So is a drug that preserves a kidney but is sold to no one who buys. Orphans do not die. They sit in the corner and wait, sometimes for sixteen years.

The clock with no owner is the last and perhaps the most important. The time a kidney spends in the cold is the single variable that every party agrees matters and that no party is accountable for. The donor hospital, the procurement organization, the airline, the courier, the transplant center and the national computer each own a slice of the hours. No one owns the hours.

The cast, in two minutes

Because the system has more acronyms than a war, here are the seven players who will appear on nearly every page.

Donor families consent, or refuse, at the bedside. Nothing begins without them.

Organ procurement organizations, called OPOs, are the fifty-five regional nonprofits that identify donors, talk to families, manage the donor's care, recover the organs and offer them to the national list. They are the front door.

Transplant centers are the hospital programs that decide whether to accept a given kidney for a given patient on their list. They are the back door, and they hold the last, fastest and least visible decision.

The Organ Procurement and Transplantation Network, the OPTN, writes the rules for sorting the list and runs the matching computer. It operates under contract to the **Health Resources and Services Administration**, HRSA, a federal agency, which has spent the past two years rebuilding it.

The Scientific Registry of Transplant Recipients, the SRTR, is the statistician: it publishes each program's results, adjusted for how sick its patients were, twice a year. Its numbers become the scoreboard.

The Centers for Medicare & Medicaid Services, CMS, pays for most dialysis and most transplants, and decides which programs and which OPOs may keep operating. When CMS speaks, boards listen.

The dialysis industry, dominated by two companies, is where almost everyone on the list spends the years while they wait. Its economics matter more than its modest share of the news would suggest.

By most accounts I found, a candidate on the list is not told, as it happens, who has turned down what on her behalf. That is not a conspiracy. It is simply how the process is built, and we will come back to it.

What this book is, and is not

It is an argument assembled from public records, peer-reviewed trials, registry reports, federal documents and reporting by journalists who did the slow work of checking. It uses sources as current as this month wherever I could find them, and it says so when the best available evidence is older. It grades its sources, openly, in the appendix, because the difference between a randomized trial and a press release is not a technicality.

It is not a history of transplantation, and it is not a patient guide. It does not tell you whether to accept a particular kidney. It does not say that the surgeon who declines is a villain, because I do not think she is. If I have done my job, you will finish the book more sympathetic to her, not less, and angrier at the structure that made her decision so easy.

It is not, finally, a document that pretends to a neutrality its author does not have. I have spent a long time on the losing side of the clock this book describes, and I have watched people I know lose to it. That is why the book is written the way it is: with the numbers first, and the people close behind. I have invented people where I needed to illustrate a pattern, and I have said so each time. Everyone I name was named in a public record.

What I am asking

Most books about health policy end with a call to awareness. I have little patience for awareness. The kidney community is aware. What it lacks is a short list of specific demands, addressed to named institutions, with numbers attached and dates beside them. The fourth part of this book is that list. It contains fifteen moves, an order in which to make them, and a plan for the first ninety days.

If you read only one chapter, read the last. If you read all of them, I hope you will arrive there angry in the useful way, the way that picks up the phone.

Denise is still waiting. So are about a hundred and forty thousand other people (Source Notes S5). Let us begin with the number.

A Map of This Book

> *"Would you tell me, please, which way I ought to go from here?" "That depends a good deal on where you want to get to," said the Cat.*
>
> Lewis Carroll, Alice's Adventures in Wonderland (1865)

This book has one destination and four stages. Knowing the route in advance will make the middle easier to follow, and it will let you skip with confidence if your time is short.

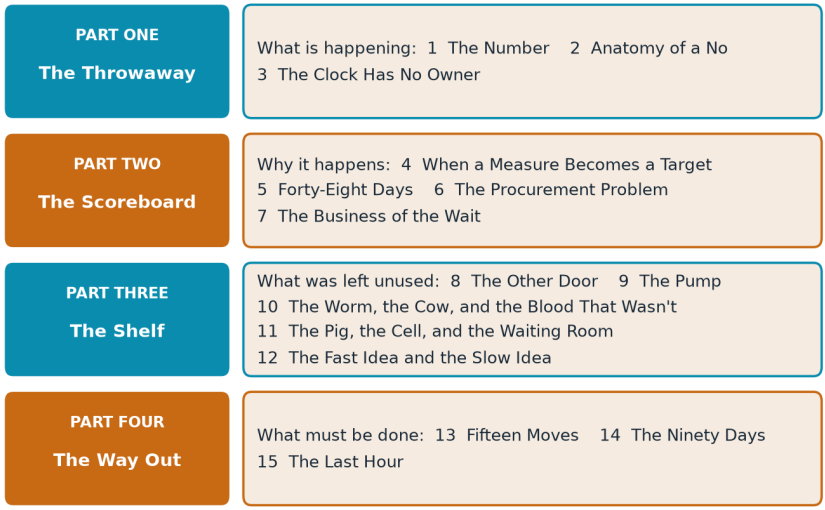


Figure 1. The structure of the book: four parts, fifteen chapters, one question answered at each stage.

The four parts

Part One, The Throwaway, asks *what is actually happening to the kidneys.* Chapter 1 puts a number on the loss and compares it with the number of people who die or become too sick to be transplanted. Chapter 2 follows a single offer through a single night, step by step, so that you can see where the decision is made and by whom. Chapter 3 explains the clock: how cold, oxygen-starved time damages a kidney, why it has grown longer, and why no one is accountable for it.

Part Two, The Scoreboard, asks *why the system behaves this way.* Chapter 4 shows how a measure becomes a target, using the federal programs that grade transplant centers. Chapter 5 tells the story of a single memo in the summer of 2025 and what it did to the national supply of kidneys in forty-eight days. Chapter 6 turns to the organ procurement organizations and the trust on which all donation depends. Chapter 7 follows the money into dialysis, the industry that earns its living while the list does not move.

Part Three, The Shelf, asks *what has already been invented, or is within reach, and has been left unused.* Chapter 8 is about the living donor, the most reliable supply we have and the one whose costs we leave to the donor. Chapter 9 is about a pump, and the sixteen years between its proof and its partial adoption. Chapter 10 is about oxygen carriers: a molecule from a cow, a molecule from a marine worm, five bankrupt companies, and a research program that has not yet earned its claims. Chapter 11 looks at the two largest possibilities on the horizon, the gene-edited pig kidney and the cell therapy that could end anti-rejection drugs, and asks what they would find when they arrived. Chapter 12 compares the ideas that spread fast with the ideas that spread slowly, including some that deserved to lose, and gives a method for telling them apart.

Part Four, The Way Out, asks *what, exactly, should be done, and by whom.* Chapter 13 lays out fifteen specific moves, each tied to an institution, a mechanism and a number. Chapter 14 turns them into a ninety-day plan for a patient, a clinic or a community group. Chapter 15, the closing chapter, steps back and asks what we owe a gift.

Each part opens with a short essay that says what the part argues, so that you can decide how closely to read it.

How each chapter is built

Every chapter has the same four parts, in the same order, so that you always know where you are.

In this chapter is a short box at the start that states the question and the answer, so that you can decide whether you need the details.

The story is a narrative, either a documented event or a clearly labeled composite, that makes the problem concrete.

The evidence sets out what is known, with the numbers, and with a plain statement of how strong each source is.

What must be done closes the chapter with the steps that follow, named by actor. Chapters 13 and 14 collect them.

Three ways to read it

If you have one evening, read the prologue, the introduction, and chapters 1, 4, 9 and 13. That is the argument in outline.

If you are a patient or a care partner, read chapters 1, 2 and 14 first. They tell you what to ask your center and how to ask it. If a friend or relative might donate, add chapter 8.

If you work in transplantation, policy or journalism, read chapters 4, 5, 9, 10 and 13. They contain the material that is least often discussed together, and the evidence most open to challenge.

Three words, defined once

Recovered means a kidney was removed from a deceased donor in order to be transplanted. *Nonuse*, the registry's term, means a recovered kidney that was not transplanted into anyone; I use *discard* for the same thing when I want to name the human act behind the statistic. *Declined* means a transplant program turned down an offer of a kidney for one of its candidates. A declined kidney can still be transplanted elsewhere, and most are. A discarded one cannot.

How to read the numbers

You will meet a great many figures. Two conventions will keep them honest.

Where a number comes from a registry, a trial or a federal document I could confirm, it is cited to a Source Note in Appendix A, graded A. Where it comes from a credible report of such a source, it is graded B. Where it comes from a single advocacy group, a manufacturer or a news release, it is graded C, and I say so in the text. The grade is a promise about how much weight a claim can bear.

Where I have done arithmetic of my own, I label it as my arithmetic and show the inputs, so that you can redo it and tell me if I am wrong. I would rather be corrected in public than flattered in private.

A glossary of the terms in this book follows the references. Skim it first if the acronyms are new.

PART ONE

THE THROWAWAY

The Question Beneath the Numbers

> *"Not everything that counts can be counted, and not everything that can be counted counts."*
>
> Attributed to William Bruce Cameron, Informal Sociology (1963)

Part One is about arithmetic, and I want to say at the outset what arithmetic can and cannot do.

It can tell you that nearly three in ten of the kidneys recovered in the United States in 2024 were never transplanted. It can tell you, by my estimate, that this is roughly 8,800 kidneys, and that in the same year 8,644 people were removed from the waiting list because they died or became too sick to be transplanted. It can tell you that a kidney's risk of failing rises with every hour in the cold. It cannot tell you, by itself, who is to blame, or whether any particular no was wrong.

That is why the part is built the way it is. The first chapter puts a number on the loss, as plainly as the sources permit, and does the arithmetic in the open so that you can check it. The second chapter slows down a single night and follows one offer through the hands that handle it, so that you can see where the decision is made and what the person making it can and cannot see. The third chapter follows the kidney itself, hour by hour, and asks who owns those hours.

I would like you to read this part with two attitudes at once. Be skeptical: not every discarded kidney was good, and the surgeon who declines is usually acting in good faith. And be curious: a figure this large, rising this steadily, in a system staffed by capable and decent people, is not an accident. It is a result. Someone, or something, is producing it. Part Two asks what.

A note on how to read the composites. In Part One you will meet a teacher named Denise and a night that I describe minute by minute. They are invented. The numbers around them are not. Each time I invent, I will tell you, and each time I count, I will cite.

CHAPTER 1

The Number

> *"The cost of a thing is the amount of what I will call life which is required to be exchanged for it, immediately or in the long run."*
>
> Henry David Thoreau, Walden (1854)

In this chapter. *Question:* how many donated kidneys does the United States fail to use, and how does that compare with the people who die waiting? *Answer:* roughly 29 in every 100, which by my arithmetic is about 8,800 kidneys a year, close to the number of people removed from the list in the same year because they died or became too sick. Not one-for-one, but too close to ignore.

I found the number in a table.

That is how most terrible numbers are found. They do not announce themselves. They sit in the overview chapter of an annual report, in the third paragraph, written by statisticians who have learned to speak of grief in the passive voice. The report is the annual data report of the national transplant registry for 2024, published in 2026, and the sentence says that the proportion of deceased-donor kidneys recovered for transplant but not transplanted rose to 29.3 percent (Source Notes S2).

I read it twice. Then I did what I would ask you to do now, which is to put it beside the year before, and the year before that, and the year the series began.

The story of a line

In 2013, 18.2 percent of recovered kidneys went unused. By 2022 the figure was 26.6 percent. In 2023 it was 27.9. In 2024 it was 29.3.

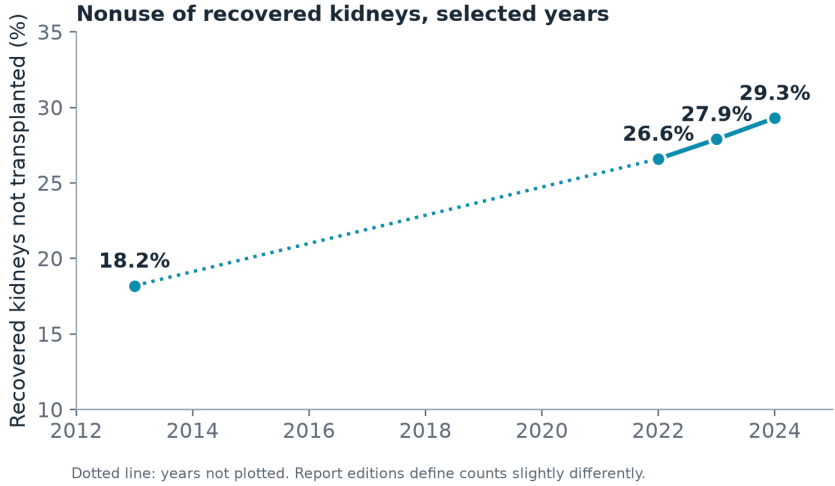


Figure 1.1. Share of recovered deceased-donor kidneys not transplanted, United States. Source: OPTN/SRTR annual data reports.

Notice what is happening around that line. Over the same decade, the system set records for everything else. A record 27,660 kidney transplants were performed in 2024, up from 27,351 the year before (Source Notes S2). A record 49,249 adults were added to the waiting list in a single year, and 143,982 adults were on it (Source Notes S6). More donors, more recoveries, more transplants, more people waiting, and a rising share of the gifts going to waste. It is the kind of pattern that looks, at first glance, like a paradox. It is not. It is what you would expect if the supply were growing faster than the system's willingness to use it.

The registry's own authors seem to read it that way. In the 2024 report they note that progress in recoveries and transplants has been offset by rising nonuse, longer cold ischemia times and higher rates of delayed graft function under broader sharing of organs (Source Notes S6). The sentence is carefully written. It does not say whose fault any of it is.

The same pattern has now reached the evening news. In February 2026 an investigative television team reported that about one in five donated organs of every kind in the United States is never transplanted, and that for kidneys the share is higher still (Source Notes S7). The registry's table and the reporter's headline are the same fact, told in two voices.

Behind every one of those unused kidneys there is a yes. A family sat in a corridor, heard the question, and answered it. Nothing in the registry records what it felt like to say yes, or what it would have felt like to learn, a year later, that the gift had been left on a shelf. That is the part of the loss that has no column.

Doing the arithmetic

The registry reports a rate. To see the loss you need a count, and the count is harder to find than it should be, so I did what I could with what I had. This is my arithmetic, and I give the inputs so that you can check it.

In 2024 there were 27,660 kidney transplants in the United States. Roughly 6,400 of them used kidneys from living donors (Source Notes S8), which leaves about 21,200 from deceased donors. If 29.3 percent of recovered kidneys went unused, then those 21,200 represent the other 70.7 percent, so about 30,000 kidneys were recovered, and about 8,800 were not used. That is roughly twenty-four a day.

I would not defend the last digit. Counts from the registry will differ a little from my estimate, because of how some multi-organ and paired recoveries are classified. A cross-check: for 2022, a report based on national data counted 7,540 unused kidneys, 26 percent of those recovered (Source Notes S9). At the rate the series was climbing, 8,800 two years later is the right order of magnitude.

Now set a second number beside it. In 2024, 3,743 candidates were removed from the kidney waiting list because they died, and another 4,901 were removed because they had become too sick to be transplanted (Source Notes S6). Together, 8,644 people.

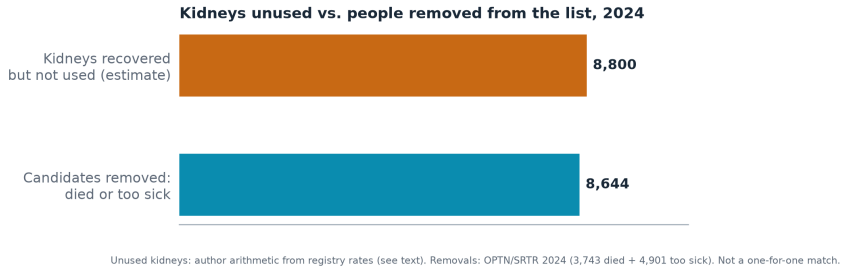


Figure 1.2. Kidneys recovered but not used (my estimate) and candidates removed from the list for death or illness, 2024.

I need to be careful here, and I want to be seen being careful. The 8,800 kidneys and the 8,644 people are not the same thing. A great many of the unused kidneys would not have suited the people who died; a great many of those people were too ill or too complicated for any kidney. Experts who reviewed the 2022 discards estimated that roughly 62 percent of the unused kidneys would have been viable, which would mean 4,675 dialysis patients who could have received one and did not (Source Notes S9). Four thousand six hundred is not eight thousand. It is also not a small number.

What the comparison does is change the scale of the question. It makes it impossible to say that the shortage is simply a matter of there not being enough kidneys. In the same year the system lost nearly as many gifts as it lost lives, and it recorded the first as a statistic and the second as a tragedy.

What a kidney is worth

There is one more number I want to put on the table, because it is the one that will matter to the people who write the checks.

According to the national dialysis registry, the per-person annual cost to traditional Medicare in 2021 was about $99,325 for a patient on in-center hemodialysis and about $43,913 for a patient with a functioning transplant (Source Notes S10). The difference is $55,412 a year.

Suppose nonuse returned to its 2013 level of 18.2 percent. Of 30,000 recovered kidneys, 5,460 would go unused instead of 8,800. That is about 3,300 more transplants a year. At the 2021 difference, that is roughly $185 million a year in

dialysis cost avoided, in a scenario where the only change is that the kidneys already recovered are put to use. I offer this as a scenario, not a forecast. It leaves out the cost of the transplants themselves, the drugs, the complications. But it does not include what matters most to the people involved, which is time. Each of those 3,300 people would be on dialysis for years less than they otherwise would be.

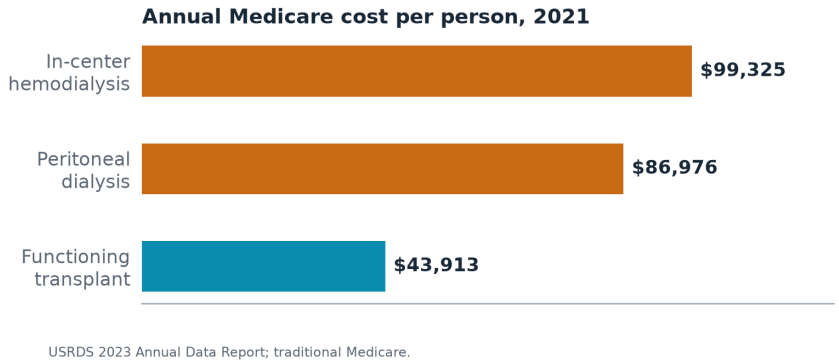


Figure 1.3. Annual cost to traditional Medicare per person, by treatment, 2021. The functioning transplant costs less than half of in-center hemodialysis.

Who gets thrown away

The kidneys that go unused are not a random sample. They have a profile, and the profile is instructive.

Kidneys from donors aged sixty-five or older went unused 68.9 percent of the time in 2024. Kidneys that had been biopsied at the time of recovery went unused 40.8 percent of the time; kidneys that had not been biopsied, 6.4 percent (Source Notes S2). That last contrast is the most striking in the whole report, and I will spend a chapter on it. A tool meant to assess quality appears to be one of the strongest predictors of waste.

The reasons recorded for declines point the same way. In one analysis of national discards, the single most common reason was that no recipient could be located, at about 35 percent; the second was biopsy findings, at about 29 percent

(Source Notes S11). The first of these means something worth pausing on. It means that the kidney was acceptable on paper, and that the machinery of offers could not find, in time, a person and a center willing to take it.

Is there a right rate?

Honest people disagree about this, and I do not know the answer.

What can be said is that the rate in the United States has not always been this high and is not this high everywhere. An analysis published in 2018, using a somewhat different definition from the registry's, found that the discard rate rose from 14.9 percent in 2000 to 19.0 percent in 2015 (Source Notes S12). A comparison of American and French practice over the years 2004 to 2014 found that the United States discarded 17.9 percent of the 156,089 deceased-donor kidneys it recovered, and France discarded 9.1 percent of its 29,984. The kidneys that Americans transplanted came from donors with an average age of about 36; the French, about 51. When the authors applied the French pattern of acceptance to American donors, they estimated that 17,435 of the kidneys discarded in the United States, 62 percent, would have been transplanted, adding an estimated 132,445 years of kidney function over the decade (Source Notes S3).

That last figure is a model, not an observation, and the French system differs from ours in many ways. But the contrast in who gets transplanted should make us curious: a country with older, riskier donors used more of its kidneys.

The median of sixteen

The last piece of evidence is the one that gave this book its first scene. In the study of about fourteen million offers, candidates who died without a transplant had received a median of sixteen offers over 651 days on the list. In the great majority of declines, about 93 percent, the centers attributed the decision to the quality of the organ or the donor. The authors cautioned that some declines were surely right and that they could not follow the fate of the transplants that followed (Source Notes S1).

I find this study more honest than most of its critics and most of its admirers. It does not say that Denise was wronged. It says that a person can wait for 651 days and be passed over sixteen times, and not once be the cause of it.

What must be done

Three things follow from the number, and each will return in Part Four.

Publish it. The nonuse rate should be reported monthly, by organ procurement organization and by transplant center, in a form a patient can read. A rate that is published once a year in a statistical report is a rate no one is accountable for.

Set a target. A national goal, with a number and a date, would turn a trend into an obligation. If the goal is to return to 18.2 percent in five years, say so. If it is something else, say that.

Count the yes. The system counts the kidneys it transplants and the graft failures it must explain. It should also count the kidneys it declines, the patients who were next in line, and what happened to them. Chapter 2 shows how that can be done.

The number is not the argument. It is the thing the argument has to explain. For that we will have to leave the registry and follow one offer through one night.

CHAPTER 2

Anatomy of a No

> *"It may seem a strange principle to enunciate as the very first requirement in a Hospital that it should do the sick no harm."*
>
> Florence Nightingale, Notes on Hospitals (1863)

In this chapter. *Question:* who decides that a kidney will go unused, and how? *Answer:* a single tired professional, in minutes, using a drop-down menu, under pressure from a scoreboard she did not design. The decision is not irrational. It is rational for her, and costly for the person next in line. Nobody records the cost.

I want to slow down a night that, in life, takes less than a minute.

What follows is a composite. I built it from the published descriptions of how offers work, from the research on how centers decide, and from the structure of the national system. Nothing in it is a transcript. Every constraint in it is real.

11:42 p.m. — The call

The offer arrives as it always does, not as a call about a person but as a record: a donor number, a blood type, an age, a set of lab values, a score. The coordinator has seen thousands. She does not read the whole screen. She reads the parts that predict the next ten minutes.

The score is the first thing. Since the allocation overhaul of 2014 the system has summarized the quality of a deceased donor's kidney in a single number, the kidney donor profile index, a percentile that runs from 0 to 100. A low number means a kidney expected to last longer than most; a high number, shorter. The

index is a statistical tool, built from the donor's age, blood pressure, kidney function, cause of death and other factors. It was revised in 2024. It is also, in practice, a sorting device: the higher it is, the more likely the kidney is to be declined.

The second thing she reads is the clock. The kidney has not yet been recovered, so the question is when it will arrive. Distance, weather, flights, a courier who may or may not be on call. The system has grown larger in recent years, and so have the journeys. Since the 2021 change that shared kidneys across wider circles around each donor hospital, the time kidneys spend in the cold has gone up, and with it the rate of what clinicians call delayed graft function, the need for dialysis in the first week after the operation. In 2023 that rate was about 26 percent (Source Notes S13).

The third thing is the biopsy. At the donor hospital, someone has cut a thin wedge from the kidney, frozen it, and sent it to a pathologist, perhaps a general pathologist who rarely reads kidneys, perhaps at two in the morning. The report says some of the filtering units, the glomeruli, look scarred.

11:44 p.m. — The screen

The coordinator now does what no algorithm does for her: she thinks about her patients.

At her center, a candidate who is sixty-three and on dialysis has been on the list for four years. Another, fifty-four, has diabetes. A third is younger and healthier but sits lower on the list. Each is a different bet. A kidney with a few warning signs is a good bet for the older patient, who might die waiting for a better one, and a worse bet for the younger, who has decades to use a long-lasting organ.

She has one more thing to consider, and it is the one no one writes in a textbook. The center's last stretch of results.

11:51 p.m. — The weighing

The surgeon, reading the same screen, is carrying a ledger she did not choose.

On one side are the consequences of a no. They are nearly invisible. The candidate does not know she was offered the kidney. The kidney goes to someone else, or does not. If the candidate dies, no record connects her death to this offer. The decline will appear, at most, as a line in a national database, a “reason” chosen from a menu.

On the other side are the consequences of a yes, and they are anything but invisible. If the kidney does not work right away, the patient may stay in the hospital longer, the program’s costs rise, and the surgeon will have to explain to a frightened family why the thing they hoped for is not yet working. If the kidney fails within the year, the failure is counted. It will appear in the published results for the program, adjusted for how risky the transplants were, and compared with a prediction. If the program falls too far below the prediction, a review follows. We will come to what a review can do in the next chapter. For now, notice only the asymmetry. This is the invisible ledger at work, in a real room, at midnight.

Nightingale’s line at the head of this chapter is one of the most famous in the history of hospital reform, and it is the right place to start. *Do no harm.* A no is the choice that cannot harm the patient on the table. It is not the same as doing no harm. It is only the choice whose harm happens somewhere else.

11:53 p.m. — The eleven seconds

She says no. The coordinator selects a reason from a drop-down list and clicks. The record is sent.

How good is the judgment behind a click like that? The research is uncomfortable. A kidney is, on average, offered about seven times before someone accepts it (Source Notes S14). In a 2017 randomized study of sixty-eight physicians shown the same offers, the variation among the decisions of doctors within a single center was roughly ten times the variation among centers. In a 2023 survey, the statistic that measures agreement among experienced reviewers looking at identical cases, called Fleiss’s kappa, was 0.13, a value that statisticians describe as slight (Source Notes S14). In plain terms: shown the same kidney, two skilled surgeons would often disagree, and the difference between them would be larger than the difference between hospitals.

Centers differ, too. A study of 193 adult programs, using offers from 2021 to 2023, found that programs accepted a median of 12.5 percent of offers of kidneys that were not considered suboptimal, and a median of 7.2 percent of suboptimal ones; the spread was widest for kidneys with more than thirty-six hours of cold time (Source Notes S15). No one believes that the patients at hospital A are three times more suited to marginal kidneys than the patients at hospital B. The difference is a difference of appetite for risk.

The biopsy at two in the morning

I said the biopsy would matter, and it does, more than almost anything else on that screen.

Procurement biopsies, the quick frozen sections read at the donor hospital, are a standard tool. They are also, according to a body of studies, poorly reproducible. In one single-center analysis, the classification of a kidney as optimal or suboptimal at procurement matched the later reading of the same transplanted kidney in only about 64 percent of cases, and for the most important feature, scarring of the glomeruli, agreement barely exceeded chance (Source Notes S16). In another, repeat biopsies of the same kidney agreed on glomerulosclerosis at a kappa of 0.12; and the second biopsy, not the first, predicted how the graft later fared. A case-control study of discards found that only one feature, glomerulosclerosis greater than 20 percent, was independently associated with discard (Source Notes S16).

And the national result: kidneys that had been biopsied went unused 40.8 percent of the time in 2024, against 6.4 percent for kidneys that had not (Source Notes S2). Some of that is selection; a surgeon biopsies the kidneys she is already worried about. But some of it, plausibly, is the biopsy itself, an imperfect measurement turned into a verdict.

What happens after a no

A no does not end the night. It starts another clock.

The kidney has been offered to the next candidate on the list. If that center also says no, to the next. Each decline takes minutes to hours; each hour is an hour of cold time. A study from the allocation system's own analysts noted that waiting for an offer to be declined can lengthen ischemic time (Source Notes S17). By the time a kidney reaches a center that would accept it, it may be too old to be worth accepting. That is the logic that produces the single largest reason for discard in one national analysis: *no recipient located*. The recipient exists. By then, the kidney no longer does.

The first fix, and what it proves

There is one reform already underway that tells us something. Offer filters are a tool that lets a program tell the system, in advance, which kinds of kidneys it will never accept for its candidates, so that those offers are not sent. In a pilot, participating centers avoided about two-thirds of the offers they would otherwise have received, and some increased their transplant volumes while receiving up to 75 percent fewer offers (Source Notes S18). The board of the national network has now approved a version in which filters are switched on by default, based on a program's own acceptance history (Source Notes S18).

I mention it for two reasons. It is evidence that the process contains a great deal of wasted motion that can be removed without harming anyone's results. And it shows the pattern this book is about: a tool that improves the system as a whole was available, and it took years to get from a pilot to a default.

What must be done

Give every candidate an offer ledger. The most important thing the night reveals is that the person whose life depends on the decision does not see it. A quarterly statement, sent to each candidate, saying how many kidneys were offered on her behalf, how many were declined, and the reasons, would let her ask the questions the system now makes unnecessary. It would not force a center to accept a kidney. It would force a center to look at what it is declining. Chapter 13 develops this idea.

Make the reasons real. A drop-down menu is not a reason. Reasons should be specific, audited and compared with what happened to the kidney afterward. If a kidney was declined for an organ-quality concern, and a different center transplanted it into a patient who did well, the first decision should be visible to the first center.

Take the second look. A kidney that has been declined more than a set number of times should be routed, automatically, to the centers with the best record of using hard-to-place kidneys. That is not a change to the clinical judgment. It is a change to the plumbing.

Denise's surgeon was not careless. She was human, rational and unobserved. The next chapter tells you what makes a kidney's hours so costly, and why no one owns them.

CHAPTER 3

The Clock Has No Owner

> *"I returned, and saw under the sun, that the race is not to the swift, nor the battle to the strong… but time and chance happeneth to them all."*
>
> Ecclesiastes 9:11 (King James Version, 1611)

In this chapter. *Question:* how much does the time a kidney spends in the cold matter, and who is responsible for it? *Answer:* it matters a great deal, in a measurable and steady way, and in a chain of six organizations, no one is responsible for the total. The most expensive variable in the system is the one with no owner.

Follow the kidney, not the offer.

Here is one imagined journey, a composite again, from the instant its donor's blood stops flowing through it to the instant a recipient's does. I have made the timeline plausible and unremarkable, not extreme. Most kidneys have a worse night than this one.

Hour	What happens	Who owns it
0	Cold fluid is flushed through the kidney; it is bagged and boxed in ice	Recovery team and procurement organization
1 to 5	Offers are made, declined, remade; the kidney is accepted by a center in another state	National matching system and the centers
5 to 7	A courier takes the box to an airport; a flight is found	Procurement organization and courier
7 to 11	The flight; a delay for weather; ground transport	Airline, courier
11 to 14	Arrival; crossmatch and preparation; the recipient is brought in	Transplant center
14 to 17	Operating room; the kidney is placed in the patient	Transplant center
17	Blood flows through the kidney for the first time in 17 hours	—
Total	Seventeen hours of cold, starved time	**No one**

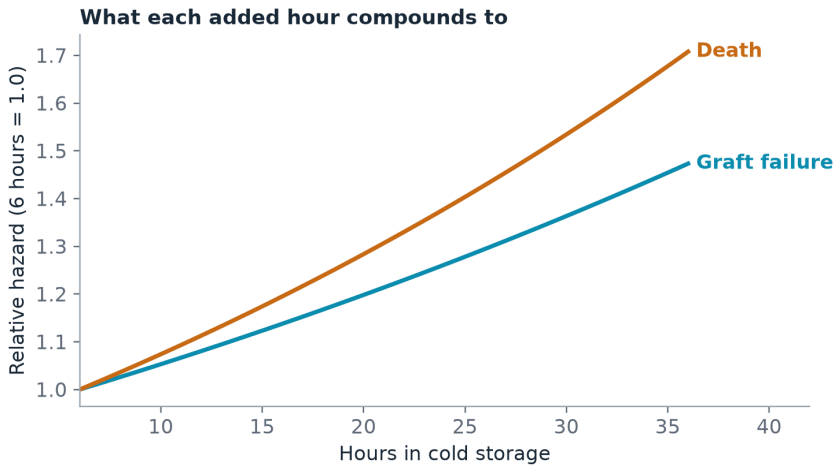
Look at the last row. It is the one that matters, and the one that has no name in the right-hand column.

What the cold does

A kidney removed from a body is not dead. It is living on what it stored. Cold slows its chemistry the way a refrigerator slows milk, but it does not stop it. Without blood flowing through it, its cells run out of oxygen and then of the molecule that carries their energy. Waste builds up. Cell membranes start to leak. And then the second injury comes, the one that surprises most people: when blood finally flows back in, the sudden return of oxygen to starved cells produces a burst of harmful chemistry that causes damage of its own. Researchers call the whole sequence ischemia-reperfusion injury, and it is the biological mechanism behind most of what follows.

The consequence is measurable in a way that is rare in medicine. A French cohort of 3,839 kidney transplants performed between 2000 and 2011 found that each additional hour of cold ischemia raised the risk of graft failure by about 1.3

percent and the risk of death by about 1.8 percent. Those are small steps. But steps compound. A kidney that spent thirty hours in the cold, compared with six, carried about 40 percent more risk of failure (Source Notes S19).



Extrapolated from published per-hour hazard ratios (1.013 and 1.018); author calculation, illustrative.

Figure 3.1. Relative risk of graft failure and of death rising with hours of cold storage, from a French cohort. Each added hour compounds the last.

A 2026 study from Freiburg, small and from a single center, looked at 74 pairs of kidneys from older donors and found that when cold time exceeded twelve hours, compared with under eight, the odds of delayed graft function were more than six times higher (an odds ratio of 6.3, with a wide margin of uncertainty), and the risk of death was roughly tripled (Source Notes S20). I would not rest a policy on a study of 74 pairs. I mention it because it points the same way as the larger studies, and because its subject is the kind of kidney, older and less than perfect, that the system is most likely to discard.

And the effect of delayed graft function is not trivial. When a transplanted kidney does not work at once, the patient needs dialysis in the first week, often stays longer in the hospital, and, in registry studies, is more likely to suffer rejection and a shorter graft life. In 2023 about 26 percent of deceased-donor kidneys in the United States went through it (Source Notes S13).

Why the hours have grown

You might expect a system to shorten the hours over time. This one has done the opposite, and the reason is instructive.

In 2021, the allocation system for kidneys was changed. Previously organs were offered first within a local area and then more widely. The new system replaced the old boundaries with circles of fixed radius around the donor hospital, to give candidates in more places a fair chance at the best kidneys. It was a response to a legitimate complaint about geography. It also meant that more kidneys traveled farther. Analyses of the change found that regional sharing added roughly 2.9 hours of cold time compared with local sharing, and that the national network's own committees acknowledged that broader sharing increases the time it takes to place an organ, the hours it spends in the cold, and the rate of discard (Source Notes S21). The authors of the 2024 national report say the same in their opening paragraphs (Source Notes S6).

Notice that no one made a mistake. A policy was designed to solve one problem, fairness across geography, and it created another, longer clocks, because the second problem had no owner to raise a hand.

The pattern is also uneven. A study of 206 transplant centers found that the median cold time for kidneys recovered at the same hospital where they were transplanted ranged from about 3.3 hours to about 29 hours across centers, and that kidneys recovered on weekends were more likely to be discarded than transplanted (Source Notes S21). Time is not a fixed fact of nature. It is a variable that some organizations manage better than others.

Six hands, no owner

It is worth naming the hands.

The donor hospital controls the pace of recovery. The procurement organization controls packaging, transport and offers. The national network's computer controls the order of the list, and the allocation rules it enforces influence how far a kidney must travel. The centers control the speed of their decisions and the readiness of their operating rooms. The airlines and couriers control the journey. And the federal agency that supervises all of them has, in

recent years, been rebuilding the machinery of oversight, but has not yet made the total number of hours for any given kidney the responsibility of any single party.

The federal government has noticed. A working group convened by the Health Resources and Services Administration produced a final report in May of 2025 with twenty recommendations on organ transport, and in 2025 the Federal Aviation Administration launched a website devoted to the air transport of organs (Source Notes S22). Industry briefings have also noted that the contract then in force did not permit the agency to require organs to be tracked in transit (Source Notes S22). These are steps in the right direction. They are also, notice, steps on one slice of the clock.

The accountability test

Here is a test I will apply to every system in this book. *If a number is important, someone's job should depend on it.*

By that test the clock fails. The national registry publishes median cold times, and the researchers who study them agree they matter. But if a kidney is lost to a six-hour delay at an airport, there is no person or organization to whom that loss is charged. If a procurement organization turns a kidney around faster than its peers, it earns nothing for it. If a transplant center delays a decision to gather a team, the cost lands on a kidney and a patient it will never see.

Contrast what happens to a number that someone does own. A program's one-year graft survival is owned by the program's surgeon and director; if it slips, a review begins. It is therefore watched, discussed, and managed. The clock is not. The difference in how the two numbers are treated is the difference between a number that has a name beside it and one that does not.

What an owner would do

Suppose we gave the clock an owner. What would change?

First, the hours would become visible. A public dashboard of cold time per kidney, by procurement organization, by center and by route, updated monthly, would show who is fast and who is slow, and why. That is not a technological challenge. The data are already collected.

Second, someone would be responsible for the total. This might be a named *organ custodian*: the procurement organization that recovered a kidney would remain accountable, in a formal sense, for its journey until it is transplanted or declined, with the authority to reroute, to arrange a faster flight, to change a plan. Today that authority is split.

Third, tools would follow. A tracked kidney is a managed kidney. A kidney that can be kept healthier during its hours, by machines that perfuse it and by additives that feed it, is a kidney whose clock matters less. Part Three is about those tools.

What must be done

Publish the clock. Cold time per kidney, per procurement organization and per center, monthly and public.

Name the owner. Make a single party formally responsible for each kidney's journey, from the moment of recovery to the moment of use or decline.

Test every rule for its effect on time. Before the next change to the allocation system, require an estimate of what it will do to the average hours in the cold, and check afterward.

A clock with no owner is a clock that will keep running. In the next part of the book we turn from what is happening to why, and the first reason is a scoreboard.

PART TWO

THE SCOREBOARD

The Hidden Rules

"Tell me how you measure me, and I will tell you how I will behave."

Eliyahu M. Goldratt, The Haystack Syndrome (1990)

Every system has two sets of rules. The first set is written down: the allocation policy, the conditions of participation, the payment schedule. The second set is the one people actually follow, and it is inferred from what happens to the people who break the first.

Part Two is about the second set. It begins with the oldest and best-documented of the hidden rules in this field: that a program is judged by the visible results of the kidneys it accepts and not by the invisible results of the kidneys it refuses. It follows that rule through a federal standard written in 2007, a corrective agreement that made programs less willing to say yes, and a metric introduced in 2023 that is meant to repair the damage and has, in my reading, a blind spot built into its center.

Then it turns to speed. A single memorandum, sent in August 2025, changed behavior across a national network within seven weeks. That is a remarkable fact about how this system listens, and it is not a fact that most people in the field have thought about. What does it mean that the system can change in seven weeks, and yet needed sixteen years to adopt a pump?

The part closes with two institutions that sit upstream and downstream of the kidney: the organ procurement organizations at the front door, whose lapses cost the public's trust in a matter of days, and the dialysis industry at the other end of the list, where the economics of the wait are written. I have tried to say, in each case, exactly what is established, what is alleged and what is unknown.

I would ask one thing of you in this part. When you read about a rule, a metric or a company, do not ask whether the people involved are good. Ask what the structure rewards. In my experience most disagreements in this field dissolve when you do.

CHAPTER 4

When a Measure Becomes a Target

> *"When a measure becomes a target, it ceases to be a good measure."*
>
> Marilyn Strathern, "'Improving Ratings': Audit in the British University System" (1997), after Charles Goodhart

In this chapter. *Question:* why would a transplant program decline a kidney it could use? *Answer:* because the system grades programs on the visible results of kidneys they accept and not on the invisible results of kidneys they refuse. We trace the rule that created the grading, the evidence that programs responded, the attempt to repair it, and the blind spot in the repair.

In August of 2016 a reporter for STAT named Casey Ross published a story whose headline said exactly what it meant: *Hospitals are throwing out organs and denying transplants to meet federal standards.* It was picked up by PBS and a dozen other outlets, and then, like most such stories, it was forgotten (Source Notes S23). I want to bring it back, because it describes the machine this book is trying to understand, and because ten years of research since have mostly confirmed it.

The story of a rule

Begin in 2007. That March, the federal government published a rule that changed how hospitals were allowed to perform transplants. Under the new rule, programs had to apply for Medicare approval and then keep showing, at each re-approval, that their results were acceptable. The yardstick was the one-year

outcome: how many of a program's transplants were still working, and how many patients were alive, twelve months after surgery, compared with a statistical prediction of how many *should* be, given how sick the patients were and how good the organs were. A program whose results fell far enough below the prediction, and stayed there, could lose its Medicare approval and with it most of its revenue (Source Notes S24).

The intention was sound, and I want to say so plainly. In the years before the rule, some programs had poor results and little accountability. A patient has a right to know whether a hospital's transplants succeed. Measuring, publishing and enforcing that were reasonable things to do.

But look at what the rule rewarded. A transplant that failed within a year would be counted against the program that performed it. A transplant that was never performed, because the program declined the kidney or declined the patient, would not be counted at all. The yardstick measured the *visible* ledger, the yes that went wrong, and said nothing about the invisible one.

STAT's story described what happened next. A transplant surgeon at UMass Memorial, Adel Bozorgzadeh, had co-authored a study showing a sharp rise in the number of patients removed from waiting lists after the federal standards took effect. The reporter described hospitals discarding organs and declining to transplant the sickest patients out of fear that a few poor outcomes would damage their ratings. Soon after the study was published, CMS changed its benchmarks to give programs more leeway to fail (Source Notes S23).

Notice the sequence. The rule produced a behavior. The behavior produced a study. The study produced a revision. The revision is the only part of the sequence that most people remember, and it was not the end of the story.

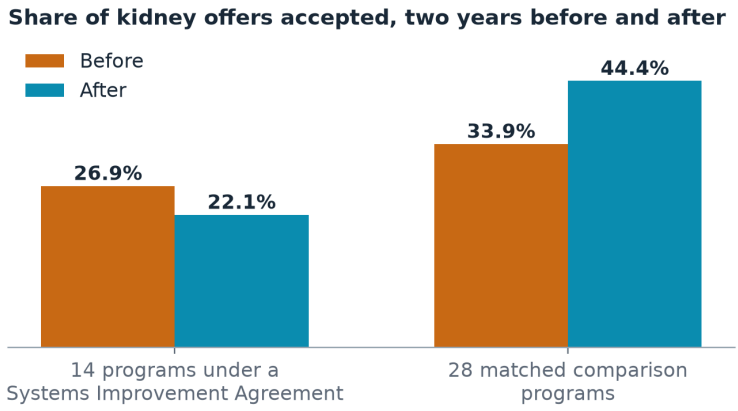
The people on the other side of the rule

To see what the scoreboard does to a program, imagine its director at the quarterly meeting.

She has seventy patients waiting. She is the person who would have to explain to her hospital's chief executive, and to its insurers, and to the federal agency, why the program's results slipped. She knows that three failures in a small year can move the number far. She knows that her program's published results appear on a public website where patients choose among hospitals, and where insurers choose which programs to include in their networks.

Now ask her what happens when she accepts a higher-risk kidney. If it works, she has performed a transplant that would not otherwise have happened; the patient lives, the result is unremarkable, and no one praises her. If it fails, she has a number to explain. She has been rational for years in a world where the downside of a yes is concentrated and the downside of a no is diffuse.

This is not speculation about her motives. It is the finding of a study of what happened when programs came under federal scrutiny. In 2018, a group at Johns Hopkins compared fourteen kidney transplant programs that had entered what is called a Systems Improvement Agreement with CMS, an agreement under which a struggling program agrees to a corrective plan, with twenty-eight matched programs that had not. Using national data from 2006 to 2015, they found that after the agreement, the programs' acceptance of kidney offers fell by 5.9 percentage points, a drop of about 22 percent. The programs accepted 26.9 percent of offers in the two years before the agreement and 22.1 percent in the two years after; the comparison programs went from 33.9 to 44.4 percent. The drop was largest for the most desirable kidneys (Source Notes S25).



Bowring et al., American Journal of Transplantation, 2018; SRTR data 2006 to 2015.

Figure 4.1. Share of kidney offers accepted by programs under a Systems Improvement Agreement and by matched comparison programs, in the two years before and after.

Read that last sentence again. The programs under pressure did not only turn down marginal kidneys. They turned down the best ones too. A program in trouble becomes cautious across the board, because caution is the only thing the scoreboard rewards. The authors warned that efforts to improve outcomes could reduce access to transplantation, and they were right.

The ratchet turns back, and the waste continues

In 2019 CMS published a rule, effective in November of that year, that removed the requirement that transplant programs meet outcome standards to keep their Medicare approval (Source Notes S26). The transplant surgeons' professional society celebrated, and said that the requirements had been so burdensome that they had led some programs to avoid transplants and discard organs, which is their claim and is not a federal finding (Source Notes S26).

You might expect the discards to fall. They rose. Nonuse was 18.2 percent in 2013. After the federal requirement was removed, it continued to climb, from the low twenties to 29.3 percent in 2024 (Source Notes S2).

I think this is the most important fact in this chapter, and it is easy to misread. It does *not* show that the 2007 rule was harmless. The Hopkins study and the reporting around it suggest it was not. What it shows is that no single rule

explains the pattern. The one-year outcome is also tracked by the national network's performance committee, which reviews programs that fall below expectations. It is published by the registry. It is used by insurers deciding whom to contract with, by hospitals deciding whom to promote, and by patients deciding where to go. Remove one pressure and the others remain. The scoreboard is not a regulation. It is a culture, and cultures do not unlearn quickly.

Trying to count the yes

To its credit, the system has begun to notice the asymmetry. In 2023, the network's performance committee began reviewing programs on a new measure: the offer acceptance ratio, which compares how many kidney offers a program accepts with how many a program with similar patients would be expected to accept. Programs with a ratio below 0.30 for adults are flagged for review (Source Notes S27). In July 2025 CMS launched a mandatory payment model, called IOTA, that scores participating transplant hospitals on three things: volume, efficiency, measured by that same offer acceptance ratio, and quality, measured by graft survival, with the three weighted 60, 20 and 20 out of a hundred (Source Notes S28). For the first time, a regulator has put a number on the yes.

I think this is the most important reform in the field in a decade, and I want it to work. But there is a blind spot in the way the ratio is built, and anyone interested in the discard problem should know it.

According to the registry's own technical documentation, offers are counted in the ratio only if they were accepted, or were declined before an offer was accepted. Offers for kidneys that were *eventually discarded* are not counted (Source Notes S27). The ratio measures the behavior of programs on the kidneys that someone, somewhere, eventually used. It is silent about the kidneys that no one did.

Think about what that means. The metric designed to reduce waste does not see the waste. A program that declines fifteen offers of a kidney that is eventually thrown away is, on this measure, indistinguishable from one that was never

asked. Until it does, the ledger remains half blind. As of this writing no results from the IOTA model have been published, and the effect of the ratio on discards has not been measured.

Whywe keep making the same mistake

The sentence at the head of this chapter has a history. The economist Charles Goodhart wrote in 1975 that any statistical regularity "will tend to collapse once pressure is placed upon it for control purposes." The social scientist Donald Campbell said it more fully four years later, in the sentence that opens this book. The anthropologist Marilyn Strathern gave it the form that stuck: when a measure becomes a target, it ceases to be a good measure.

All three were describing the same thing, and none was describing transplantation. They were describing universities, schools, police departments, and the strange fact that a social indicator can be accurate only until someone's job depends on it. The lesson they drew was not that measurement is bad. It was that every measure creates a shadow, and that you must keep looking at the shadow.

The transplant system looked at the light, the number it could count, and forgot the shadow, the lives it could not.

What must be done

Count the shadow. Every program's offer acceptance ratio should include offers for organs that were eventually discarded, and the national registry should publish, for each program, the number of kidneys it declined that were later transplanted elsewhere with good results. That is the closest thing to an audit of whether the no was right.

Risk-adjust honestly. The expected-outcome formula should account for cold time, delayed graft function and whether a kidney was machine-perfused. A program that uses a hard-to-place kidney well should be credited for the risk.

Pair every penalty with an incentive. A review triggered by a poor one-year outcome should be matched by a review triggered by a poor yield: programs that accept very few kidneys relative to their peers, given the same patients and the same offers.

Put an expiry date on every rule. Each metric and each memorandum should sunset in three years unless re-justified with evidence of its effect. The ratchet only turns one way because no one has to defend a guard once it is installed.

Test regulation like a drug. Before a change in oversight is imposed, state in advance what outcomes it is expected to produce, including an effect on the number of transplants, and measure afterward. The next chapter shows what happens when that is not done.

CHAPTER 5

Forty-Eight Days

"Early and provident fear is the mother of safety."

Edmund Burke, Speech on the Petition of the Unitarian Society (1792)

In this chapter. *Question:* how fast can the system change behavior, and in which direction? *Answer:* in under seven weeks, on the strength of an email that threatened discipline, and in the direction of fewer transplants. The same system has needed sixteen years to adopt a pump. Fear travels faster than benefit.

On August 6, 2025, an email went out to the people who run the American organ transplant network. It was a memorandum, not a rule. It changed no law, no regulation, no policy approved by the network's board. It said, in the careful language of federal oversight, that the practice of placing organs out of the order in which a computer had ranked the candidates was a concern, and that it could lead to disciplinary action (Source Notes S29).

Forty-eight days later, researchers at New York University looked at what had happened.

The story of a memo

To understand why a memorandum could matter, you need to know about a practice with an unhelpful name.

When a kidney is offered to the person at the top of the list and declined, and then to the next, and the next, a point may come at which a procurement organization, watching the clock, decides to stop working down the list and call a

center it knows will say yes. The practice is called allocation out of sequence. It can be a rescue. A kidney that has been turned down a dozen times, from a donor with a score that makes most programs hesitate, may find its only chance of use in a center that has the surgeons, the machine perfusion and the willingness to take it. The practice grew rapidly. By one count, kidneys placed out of sequence unilaterally by procurement organizations rose seventeen-fold between 2020 and early 2026 (Source Notes S30).

It can also be a problem. The match run exists because it embodies a promise: that the person who has waited longest, or who is sickest, or whose tissue fits best, will be offered an organ before someone further down. Every kidney placed out of order is a kidney offered to someone who was, by the rules, not next. If the practice is applied inconsistently, it can favor some centers and some patients over others, and it can be done in ways no one outside the process can see. The federal agency that oversees the network worried about exactly that, and it was right to.

So the memo said, in effect: follow the list.

Imagine the morning it arrived. A procurement organization's on-call team opens the email over coffee. They know the practice has been tolerated for years. They know that the last time a federal agency used the word *discipline* about a procurement organization, it was followed by a review, a corrective plan and a story in the local press. They do not know whether the memo means what it says. They know that the cost of misreading it falls on them.

Forty-eight days

A research group at NYU led by Syed Husain compared the period from August 6 to September 23, 2025, with the same weeks of the previous year and with other stretches of 2025. They found that kidney recoveries, kidney transplants and out-of-sequence allocations all fell after the memorandum. If the drop continued, they estimated, the country would perform roughly 1,500 fewer kidney transplants a year than in recent reference periods (Source Notes S29).

A separate analysis, summarized by an advocacy group for donation science, reported that the number of kidneys recovered each day fell from about 85 to about 76, and that discard rates did not rise (Source Notes S29). That detail is worth a pause. If discards did not rise and recoveries fell, then the missing kidneys were not recovered and thrown away. They were never recovered at all. Procurement teams, facing the risk of discipline, appear to have declined to take on the hardest placements.

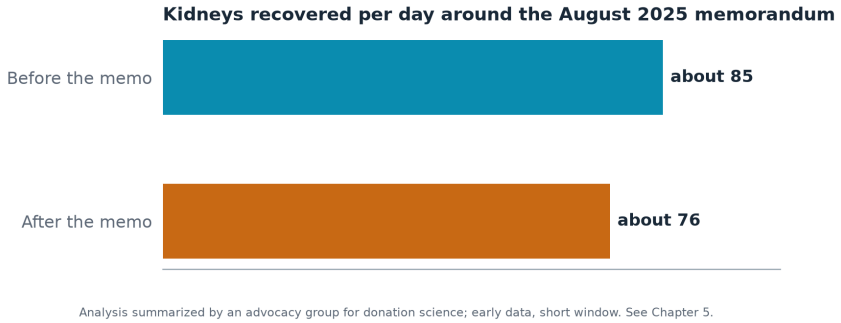


Figure 5.1. Kidneys recovered per day before and after the August 2025 memorandum, from one early analysis.

A third study in the *American Journal of Transplantation*, with a title that deserves to be quoted in full, found that eliminating out-of-sequence allocation for thirty days decreased the use of hard-to-place kidneys, in a paper called "There is nothing equitable about a wasted organ" (Source Notes S29). Three groups, three methods, one direction.

I need to be careful with the numbers, because these are early analyses, some are abstracts or trade summaries, and one is a short window. The estimate of 1,500 is a projection if the drop were to persist. I do not know whether it persisted. What I am confident about is the shape: a federal signal, delivered cheaply, produced a measurable change in a national system within weeks.

The asymmetry of speeds

Put the forty-eight days beside the other timelines in this book.

A randomized trial showed in 2009 that a pump kept kidneys working longer. By 2023 the pump was in use for about 39 percent of kidneys (Source Notes S31). Using kidneys from donors exposed to hepatitis C, made possible by new antiviral drugs and tested in a ten-patient trial in 2018, moved more slowly than the first reports suggested: in 2024 the nonuse rate for kidneys from donors with antibodies but no detectable virus was still 32.2 percent (Source Notes S32). The largest trial of hemodiafiltration, which found fewer deaths, appeared in 2023, and American patients are only now beginning to receive it (Source Notes S33).

Against these, the memo: one email, no vote, no public comment, no evidence review, forty-eight days.

This is the central asymmetry of Part Two, and it applies far beyond transplantation. A threat of punishment reaches every desk in a system in a day. A promise of benefit has to be believed, funded, trained for, scheduled and billed before it moves anyone. The system's reflexes are fast in the direction of caution, and slow in the direction of change. If you want to understand why a good idea sits on a shelf, start by asking how fast a bad one could be installed.

The case on the other side

The agency has a duty to protect the integrity of the list. Candidates who have waited for years have a right to expect that the order will be honored. Out-of-sequence placement, done in the dark and unevenly, can undermine that trust, and trust is the thing every other part of this system depends on. The federal agency has also, in the last two years, been working to rebuild oversight of a network that had been criticized for decades, including by Congress. Seen from the agency's desk, the memo was a reasonable attempt to restore discipline to a practice that had grown fast and quietly.

And I do not know what happened next. In April 2026 the network's board heard an update from a working group that had drafted a definition of what counts as an offer, criteria for an expedited placement pathway, and a framework for auditing it. The board took no action on the proposals at that meeting, and several projects remained paused pending federal review (Source Notes S34). I have not found a report of how the question was settled by the autumn.

What I can say is that an expedited pathway with a clear definition and a public audit would solve both problems at once: it would keep the list honest and it would keep hard-to-place kidneys moving. The memo got the first without the second. It was a good instinct delivered through a bad instrument.

What the chapter proves

It proves something narrower and more useful than that regulators are careless: the system is *responsive*. Within seven weeks, a national network of fifty-five procurement organizations and nearly two hundred kidney transplant programs altered what it did. A system that can change in seven weeks can change. The question is whether we will ever point that speed in a better direction.

What must be done

Assess before you send. Before any federal memorandum, rule or policy touching organ placement is issued, the agency should publish, in a page, what it expects the effect to be on the number of kidneys recovered and transplanted, and commit to a thirty-day check against that expectation. A rule that could cost 1,500 transplants a year deserves a forecast.

Replace the memo with a pathway. Define an expedited placement route for kidneys that have been declined a set number of times, with clear criteria, real-time logging and a public audit. Fairness and utilization are not opposites. They are two requirements of the same design.

Publish rapid evaluations. When a signal changes the system, the data on its effect should be available in weeks and not years. Husain's group did the work in less than two months, with public data. The agency can do the same.

The scoreboard and the memo are two of the forces that shape what happens to a kidney. The third is the organization that decides which kidneys exist at all.

CHAPTER 6

The Procurement Problem

> *"But he that filches from me my good name / Robs me of that which not enriches him, / And makes me poor indeed."*
>
> William Shakespeare, Othello, act 3, scene 3 (c. 1603)

In this chapter. *Question:* what happens to the supply of kidneys when the organizations that recover them lose the public's trust? *Answer:* the supply shrinks fast, and not only for the organizations at fault. Procurement is the front door of the whole system, and the evidence of the past two years is that the front door has been left unwatched.

In the last week of October 2024, the Associated Press reported a number that I think is the most honest measure of public trust ever published about organ donation. In the days after news coverage of a man in Kentucky who had been mistaken for dead, an average of 170 people a day removed themselves from the national donor registry. In the same week of 2023 the number had been about one-tenth of that (Source Notes S35).

One hundred and seventy a day. They did not write letters. They did not march. They opened a website, clicked a box and took their names off a list. The head of the national association of procurement organizations told the reporter that trust of that kind "takes years to regain."

The story of a pause

The case behind the headline is documented, and its details are worse than the headline.

It involved an organization in Kentucky, one of the fifty-five regional nonprofits that make up the front door of the American transplant system. A procurement organization's job is to find potential donors in the hospitals of its region, to talk to families, to manage the donor's care between death and recovery, to recover the organs and to offer them to the national list. It is a difficult, delicate and underpaid job, done in the worst hours of other people's lives.

In March of 2025 the Health Resources and Services Administration, the federal agency that oversees the network, released a report on the Kentucky organization. It reviewed about 350 cases, 351 in the agency's own summary, in which a family had authorized donation after circulatory death but the recovery had not been completed. In 73 of them, the agency concluded that the process should have been reconsidered sooner, because the patient's neurological status was high or improving. In 20.8 percent, the patient had features not conducive to this kind of donation at the initial or a later assessment. In at least 28 cases, 8.0 percent, no cardiac time of death had been recorded, which the report said suggested the patient might have survived to leave the hospital. The department's headline figure was that concerns were found in 103 of the 351 cases (Source Notes S36).

The organization disputed the characterizations. By then it had merged with a neighbor in Ohio, in October 2024, into a new entity, and the federal agency had directed the network to reopen a case that its own peer-review committee had closed a few days earlier. The corrective actions announced included a root-cause analysis, a rule that any staff member could halt a donation, and a requirement that all pauses for safety concerns be reported to the network (Source Notes S36). The agency said it had received reports of similar patterns at other organizations.

I want to be fair to everyone in this story, as the sources allow. The organization says its practices were lawful and its staff were trying to honor families' wishes. Many procurement professionals I have read say that the media

coverage has distorted a practice that, done properly, saves lives. I do not know how many of the concerns in the 103 cases would withstand full clinical review, and the agency's own phrase is *concerns*, not *violations*. But nothing in the documents suggests that the registry's 170 departures a day were an overreaction to a mere rumor.

The first decertification

In September of 2025 the federal government did something it had not done before. It decertified an organ procurement organization in the middle of a certification cycle. The target was the organization serving the region around the University of Miami. The department cited unsafe practices, poor training, chronic underperformance, understaffing and paperwork errors, and the organization said it would not appeal (Source Notes S37). Separate congressional inquiries into other allegations are, so far as I can find, allegations only.

Then there is the larger machinery. Under a rule finalized in 2020, procurement organizations are sorted every cycle into three tiers according to two measures, how many donors they produce relative to the number of eligible deaths in their region, and how many organs from those donors are transplanted. The top quarter, Tier 1, is recertified automatically. The middle compete for their regions. The bottom, Tier 3, is to be decertified. The first cycle under this system ended in 2026, using 2024 data. In January 2026 CMS proposed a further rule to decertify any organization that is in Tier 3 in all its service areas (Source Notes S38). I could find no published roster of the organizations expected to be decertified, and the process, as of this writing, is not finished.

One more study belongs here, because it asks a question that is easy to ignore. Researchers examining the organizations projected to fall into Tier 3 found that they serve populations that are disproportionately non-white. The metric that decides which organizations survive may fall hardest where the need is greatest (Source Notes S39).

What the front door has to do with the back door

Why does a book about discarded kidneys have a chapter about procurement?

Because trust is the raw material. Every kidney this book discusses began as a family's yes. If the number of people who say yes falls, there are fewer kidneys to discard and fewer to transplant. The 170-a-day figure is not a news story. It is a supply forecast.

And because procurement is where the scoreboard of Chapter 4 meets the clock of Chapter 3. An organization judged on organs transplanted, and not on organs recovered and then wasted, has every reason to maximize the first; it has fewer reasons to ask whether the kidneys it recovers can be placed. The organization that recovers a kidney bears the cost of recovery, the risk of reputational damage if something goes wrong, and, as we saw in the last chapter, the cost of a federal memorandum if it tries to place a hard kidney outside the sequence. It has no scoreboard line for a kidney it saved.

I do not claim that the metrics caused what happened in Kentucky. I found no evidence of that. I claim that a system that measures only the end of the chain, and that has shown, repeatedly, that it responds to what it measures, should expect trouble at the beginning.

What the other front doors look like

It is possible to build a system in which families are treated differently. In Spain, which has led the world in donation for three decades, the national organization places a physician coordinator in each hospital whose job is to identify potential donors and speak with families. In 2025 Spain reported 51.9 deceased donors per million people, close to the 52.6 it had reached the year before, and roughly 6,300 transplants (Source Notes S40). The comparison with the United States must be handled with care, because definitions, laws and cultures differ. But the lesson that most Spanish-model scholars draw is not about technology. It is about people placed where the families are, with time and training, and with the national organization on the hook for the result.

The research on families suggests why this matters. In an Australian interview study of 49 potential donor families, no family that had consented regretted the decision, while some that had declined did. Dissatisfaction with how hospital

staff behaved was more likely to trouble a family than the donation itself (Source Notes S41). What hurts, in the long run, is not the yes. It is the manner in which the yes is received, and the silence afterward.

What must be done

Install an independent safety officer in every procurement organization, with authority to halt a recovery and a direct line to the federal agency that cannot be overridden by the organization's own leadership. The agency has begun to require something like this. It should be permanent and universal.

Publish the pause log. Every halted donation and the reason for it should appear in a quarterly public report. Pauses are not scandals. They are evidence that the system is working. A system that does not pause cannot be trusted.

Close the loop with families. Within thirty days of a donation, every donor family should receive a written account of what became of each organ. The family that said yes should not learn about the outcome from a newspaper. This is also the best defense against the next 170-a-day week.

Measure trust. The registry opt-out rate should be tracked weekly by region and reported alongside the performance tiers. It is the only metric that tells the system, quickly, whether it has gone too far.

Count what is wasted. Procurement organizations should be graded on kidneys recovered and then not used, not only on kidneys transplanted, so that the organization that recovers a kidney has a stake in whether it finds a body.

A procurement organization's good name is, in Shakespeare's phrase, the one possession that enriches no thief. It also underwrites everything else in this book. The next chapter follows the money away from the hospital, and into the building where most of the people on the list actually live.

CHAPTER 7

The Business of the Wait

> *"It is difficult to get a man to understand something, when his salary depends upon his not understanding it."*
>
> Upton Sinclair, I, Candidate for Governor: And How I Got Licked (1935)

In this chapter. *Question:* does the economics of dialysis help or hinder the movement of patients to transplant? *Answer:* the evidence suggests it hinders, mildly and structurally, not by conspiracy but by arithmetic. A market dominated by two companies is paid for keeping the chair full. The evidence for motive is thin; the evidence for pattern is not.

Every person on the transplant list is, for the years they wait, a customer.

I mean that without cynicism. The great majority of the 143,982 adults on the kidney waiting list in 2024 (Source Notes S6) were on dialysis, in a chair, three afternoons a week. The chair belongs to a business. The business is paid by the treatment. That business has, over the last two decades, consolidated into one of the most concentrated markets in American medicine, and its incentives are one of the quieter forces shaping how fast patients move from the chair to the list and from the list to a kidney.

I want to treat this chapter carefully, because it is the one in which it is easiest to say more than the evidence allows. The people who run dialysis units are not villains. Many are nurses and technicians who know their patients' dogs by name. The companies employ tens of thousands of people, and they provide,

most of the time, a treatment that works. But a book about the invisible ledger cannot ignore a place where money, accountability and the length of the wait meet.

The story of a chair

Consider the unit, not the company. A dialysis chair is capital. The machine, the nurse, the technician, the water treatment system and the lease are fixed costs that must be paid whether the chair is occupied or not. A chair that is full three shifts a day, six days a week, is a chair that earns its keep. A patient who leaves, by transplant, by moving, or by death, creates a vacancy, and a vacancy costs money until a new patient fills it.

This is not a secret and is not unique to dialysis. It is the economics of any facility. What makes it consequential here is that the patient who leaves by transplant is leaving for a better life, and that the decision about whether and when that happens is shaped, in part, by people who work in the building the patient is leaving. The facility is the place where a patient is first told about transplantation, referred for evaluation, or not.

Do they tell? A national analysis of patients who began dialysis between 2007 and 2019, using forms that clinics submit when a patient starts treatment, found that about 15 percent were not told about transplantation at all. Patients who were informed reached the waiting list sooner. Facilities owned by large chains were more likely to inform patients, but their patients were not more likely to be waitlisted. And independent facilities that were acquired by chains became more likely to inform patients but less likely to waitlist them after the acquisition (Source Notes S42).

Hold that last finding for a moment. It does not show that chains are hiding transplantation. It shows that the knowledge and the result are not connected. Telling is cheap. Referring, evaluating and listing take staff, follow-up and effort, and they remove a chair's occupant at the end.

What the numbers say

The best-known study of the question was published in the *Journal of the American Medical Association* in 2019. Researchers examined 1,478,564 patients treated at 6,511 facilities between 2000 and 2016. They found that for-profit facilities had significantly lower rates of placing patients on the waiting list than nonprofit ones. The study's senior author, Rachel Patzer, said in the coverage that the pattern was consistent with the incentives: facilities earn more when patients stay on dialysis (Source Notes S43). I will not quote the secondary numbers from that study, which I could not check. The headline is well established.

I should say what the study does not show. It shows association and not motive. A for-profit facility may serve patients who are poorer, sicker or farther from a transplant center. The authors adjusted for many such factors. But no adjustment can separate what an organization intends from what its incentives produce.

Other evidence points in the same direction. A 2026 study of how patients arrive at dialysis found that more than half started treatment without having had care from a kidney specialist beforehand or a permanent access in place, and that these unprepared patients were more likely to be at for-profit and smaller facilities, to be male, younger, Black or Hispanic, uninsured, not informed about transplant, and living in high-poverty neighborhoods (Source Notes S44). A federal payment experiment designed to encourage transplant referral, the ESRD Treatment Choices model, which applied to roughly a third of the country from 2021, reported through three years no measurable impact on waitlisting or on living-donor transplantation (Source Notes S45).

Put these together and a pattern emerges. The incentive to refer exists in theory; in practice it has not moved the needle. The payment design was not strong enough, or the effort required was too great, or the incentives for the opposite were not weak enough.

A market with two heads

The conversation about incentives would be different in a competitive market, and this one is not.

A peer-reviewed analysis found that the two largest dialysis companies together held 59.1 percent of the market in 2005 and 77.1 percent in 2019 (Source Notes S46). Other estimates put their combined share anywhere from about 72 to 80 percent. Adam Smith, in *The Wealth of Nations* in 1776, wrote that "people of the same trade seldom meet together, even for merriment and diversion, but the conversation ends in a conspiracy against the public, or in some contrivance to raise prices." Smith was not describing dialysis. He was describing the natural tendency of concentrated trades. The most one can say is that the dialysis market has the structure he warned about.

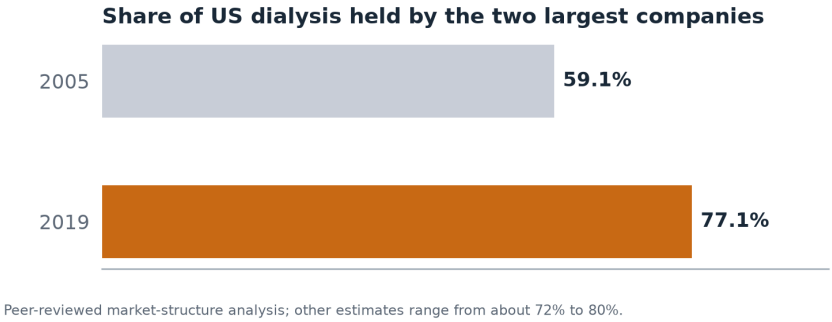


Figure 7.1. Combined share of US dialysis held by the two largest companies, 2005 and 2019.

The industry's finances are contested. The Medicare Payment Advisory Commission reported that the aggregate Medicare margin for outpatient dialysis was negative 0.2 percent in 2023 and 4.5 percent in 2024, and recommended that Congress eliminate the payment update for 2027. An industry trade group has argued that the commission's figure overstates the margin by roughly two points, and that the average conceals many struggling clinics (Source Notes S47). Both statements may be true. A modest Medicare margin can coexist with a much more profitable commercial insurance business, which is where the next part of the story goes.

What the courts and the regulators have said

I will separate what is established from what is alleged, because the difference matters.

The established part is that the largest company entered a corporate integrity agreement with the federal government in 2014, as part of a resolution of allegations about arrangements with physicians. In July 2024 the Department of Justice announced a settlement of more than $34 million of allegations of illegal kickbacks, a fact I found referenced in a journal article and not on the Department's own page (Source Notes S48).

The alleged part is larger. A 2016 whistleblower alleged that the two largest companies steered patients toward a charity's premium assistance program, which pays insurance premiums for patients, in ways that moved them from Medicare onto private plans that pay far more per treatment; the charity, whose funding comes largely from the two companies by advocacy accounts, disputes these characterizations. California passed a law to cap payment for such patients, and a court enjoined it in 2019 (Source Notes S48). I did not find the resolution of the whistleblower's suit, and I do not claim it was proven.

A related allegation concerns the procedures that keep a dialysis patient's access working. In 2022 the Department of Justice sued a vascular access subsidiary of the second company, alleging that at least 1,288 of 2,303 angioplasty procedures among sixty patients, 56 percent, were medically unnecessary (Source Notes S49). It is an allegation. A predecessor company paid $3.59 million to resolve a similar False Claims Act matter. And industry advocates argue, with some evidence, that patients treated at independent access centers have more fistulas and fewer central venous catheters than those treated in hospitals (Source Notes S49).

I do not draw a conclusion from this list. I draw a pattern: a business in which the decisions that make the most money are the ones least visible to patients, and in which enforcement arrives years later, if at all.

What the machines are not doing

There is a second way to see the business: by what its machines do not do.

In 2023, a large European trial of high-dose hemodiafiltration, a treatment that cleans blood using convection as well as diffusion, found that patients lived longer than on the standard high-flux hemodialysis the American system uses. Among 1,360 patients followed a median of thirty months, 17.3 percent of those treated with hemodiafiltration died, against 21.9 percent of those on standard dialysis, a hazard ratio of 0.77 (Source Notes S33). The benefit seemed to come mainly from fewer deaths from infection. The trial has critics, including about the patients selected.

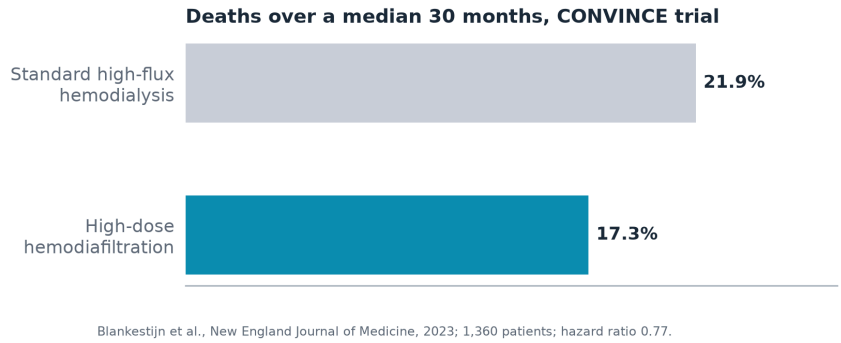


Figure 7.2. Deaths over a median of thirty months in the CONVINCE trial of hemodiafiltration versus standard hemodialysis.

American patients cannot easily receive this therapy. The leading manufacturer received regulatory clearance for a machine capable of it in February 2024, and for an updated version in May 2025, with broader commercial use planned for 2026 (Source Notes S33). The company says roughly 160,000 in-center machines in the country could eventually be replaced. Whether the American industry will adopt it faster than it adopted home dialysis, whose share of new patients rose only from 6.8 percent in 2010 to 13.3 percent in 2020 (Source Notes S50), remains to be seen.

I cannot claim a motive for the delay. I can note that a treatment that costs more to deliver and does not increase the payment per treatment is a hard sell inside a bundle that pays a fixed amount.

What must be done

Make referral the default. A randomized trial across 134 dialysis facilities in Georgia, covering more than 9,000 patients, found that a structured intervention improved referral for transplant evaluation and narrowed the racial gap in referral (Source Notes S51). That intervention should be the standard of care, funded and required in every facility, with referral to evaluation within ninety days of starting dialysis.

Put the number on the first screen. Every dialysis facility's rates of patient education, referral, evaluation and waitlisting should be displayed in the same place and the same size as its star rating, so that patients and families see it before they choose.

Open the books. The compensation of medical directors, the ownership stakes of physicians in vascular access centers and in dialysis facilities, and the relationships between charities and the companies that fund them should be disclosed in a public, searchable database.

Ask the antitrust question. A market in which two firms hold more than three-quarters of the business has earned an independent review by competition authorities. A review is not an accusation. It is due diligence on behalf of patients who cannot shop.

Pay for what works. Hemodiafiltration, home dialysis and early referral should be paid for in a way that makes them attractive to deliver, with a requirement that results be tracked in a national registry, so that whatever is learned, the system learns it quickly.

A kidney patient is not a customer. For years she is treated as one, and nothing in the economics of the wait makes the next chair her priority. The next part of the book looks at what has been invented, and left on the shelf, in the meantime.

PART THREE

THE SHELF

What Was Left Behind

> *"The future is already here; it is just not evenly distributed."*
>
> Attributed to William Gibson (c. 1993)

A shelf is where you put what you intend to use later.

Part Three is about the things this field has invented, tested and put on the shelf, and about the reasons they are still there. It takes four of them, in rough order of how well the evidence supports them.

The first is not a machine at all. It is a person: the living donor, whose yes has been the most reliable source of kidneys for half a century and whose cost the system has, until very recently, left her to bear. The second is a pump, with a randomized trial and a ten-year follow-up behind it, used for fewer than four kidneys in ten. The third is a family of oxygen-carrying molecules, one borrowed from a cow and one from a marine worm, whose history includes five failed companies and whose current evidence is early and partly company-sourced. The fourth is a pair of futures, a gene-edited pig kidney and a cell therapy that could end the need for anti-rejection drugs, each of which would remove an old constraint and each of which would walk into the structure of Part Two.

The last chapter in this part steps back and asks why some good ideas spread in months and others in generations. It also tells the story of an idea that deserved to lose, because a book about slow adoption that never admits a bad

idea was adopted too fast is only advertising.

The reader should note a difference between this part and the one before it. In Part Two, I was describing behavior. Here I am describing evidence, and the standards are different. Where a claim rests on a randomized trial I say so. Where it rests on a company's press release, I say that too. If you keep one rule in mind, let it be this: a proven tool that goes unused and an unproven tool that is hyped are two different failures, and the cure for each is the same experiment.

CHAPTER 8

The Other Door

> *"Greater love hath no man than this, that a man lay down his life for his friends."*
>
> John 15:13 (King James Version, 1611)

In this chapter. *Question:* if the front door of the system, the deceased donor, leaks, what about the other door, the living donor? *Answer:* it is the most reliable supply we have, it has not grown back to its 2019 level, and the reason is partly that we ask the person who gives a kidney to carry costs that the program saving the most money does not. It is the invisible ledger again, with a human name on it.

Meet a man I will call Teodoro. He is invented, and everything about his situation is ordinary.

He is forty-one and he wires houses. He is paid by the hour, and he has no paid leave. His younger brother, thirty-seven, is on dialysis, has been for two years, and has been told by a transplant center that the quickest route to a new kidney is a living donor. Teodoro is a match. He is also, as the evaluating team's financial coordinator explains to him, kindly and without expecting him to be surprised, going to need about six weeks away from work after the surgery, during which no one will pay him. The center will pay for the evaluation and the operation, as the law and custom require, but not for the gas to drive three hours each way for the appointments, not for the motel, not for his lost wages, not for the child care when his wife drives him, not for the rent.

Teodoro does the arithmetic on the drive home. He is a good brother. He will do it. And as he does the sum, he understands something that is invisible to the system he is about to enter: that he is about to save a government program about fifty-five thousand dollars a year, for as many years as his brother's kidney lasts, by paying out of his own pocket for the privilege.

I want to take a chapter to look at this door, because it is the one supply of kidneys that does not depend on a tragedy, a clock, a pump, a regulator or a memo. It depends on one person's yes. And because it is the cleanest illustration in the book of what happens when the benefit of a good thing lands on one party and the cost on another.

What a living kidney is worth

A kidney from a living donor tends to outlast one from a deceased donor, and the registry numbers are striking: in one analysis, the median survival of a transplanted kidney from a living donor was about 19.2 years in recent transplants, against 11.7 years for deceased-donor kidneys (Source Notes S52). A living-donor transplant is scheduled. The cold time is minutes. The surgery can happen when the recipient is well, before the years of dialysis have taken their toll.

For the payer the arithmetic is as plain as I have given it elsewhere. In 2021 traditional Medicare spent about $99,325 per person on in-center hemodialysis and about $43,913 on a person with a functioning transplant, a difference of $55,412 a year (Source Notes S10). Every living-donor transplant that lasts the median span returns that difference for years.

Now ask who pays. In the American system, by custom and by law, the recipient's insurance covers the donor's medical evaluation and surgery. What it does not cover is nearly everything else, and the person who absorbs that is the donor.

The number that has not recovered

Here is the count. A news report that cites national registry figures puts the number of living kidney donors in the United States at 6,290 in 2023, 6,419 in 2024 and 6,522 in 2025, against 6,867 in 2019 (Source Notes S53). I give this as a single-source figure, and a registry table should replace it. If it is right, living donation in 2025 was still about 5 percent below where it stood before the pandemic, in a country whose waiting list had grown.

It is also worth stating what long-run analyses have found. Living kidney donation in the United States fell after a peak in the middle of the 2000s, with the drop largest among men, Black adults and younger and lower-income people (Source Notes S54). A 2026 article in the *Medical Journal of Australia* found 253 living-donor kidney transplants in 2024, against 354 in 2008, with lower socioeconomic groups disadvantaged (Source Notes S54). The pattern is not American alone. And it is not what you would expect of a pure act of generosity, which should not depend on one's income.

That pattern is the signature of a cost. When a gift is made harder for the poor than for the comfortable, the thing that is being measured is not generosity. It is the price of generosity.

What the law just did

In February of 2026 Congress did something about it. The Honor Our Living Donors Act, enacted as part of the Consolidated Appropriations Act, 2026 (Public Law 119-75) on February 3, directs the federal living-donor reimbursement program to stop considering the recipient's household income in deciding whether a donor qualifies, and to reimburse donors for travel, lost wages, and child and elder care, up to a ceiling of $6,000 per organ, with a duty to report annually to Congress. The agency published a request for public comment on its approach in July 2026. In the donor-side test that remains, priority in the proposal goes to donors at or below 350 percent of the poverty guidelines (Source Notes S55).

I regard this as one of the most important pieces of transplant legislation of the decade, and I want it to work. I also want to point to two details in the sources. One, a letter from a physicians' group reported that roughly 21 percent of donors reach the $6,000 cap, which means that for one donor in five the ceiling does not cover the cost. Two, the program is a reimbursement, which means the donor must front the money and wait.

Private programs have grown up to fill part of the gap. The National Kidney Registry and the American Kidney Fund describe programs that offer wage reimbursement and life-insurance protection to some donors (Source Notes S56). The American Kidney Fund's May 2026 report card on state protections for living donors found gains in several states and little or no progress in many (Source Notes S56). These are welcome. They are also a reminder that for years the cost of a gift was handled as an act of charity by nonprofits, not as an obligation of the program that saves the money.

A bill pending in the House would go further. The End Kidney Deaths Act, introduced in April 2025, proposes a refundable tax credit for people who donate a kidney to a stranger, $10,000 in the year of donation and in each of the following four years, for $50,000 in all, applying to kidneys removed after 2026 and expiring in 2036. It had about sixty cosponsors by one campaign's count in March 2026, and a policy tracker rated its chance of passage low (Source Notes S57). Its supporters say it would increase donation and cut dialysis costs. Its critics worry about the ethics of financial incentives for an organ that federal law forbids buying. I am not endorsing or condemning it. I raise it because the argument it provokes is the right one: the present arrangement, in which the donor's generosity is rewarded with a bill, cannot be right.

The chain

There is one innovation in this part of the system that has worked, and it deserves its due.

The idea of paired exchange is simple to state: two donor-recipient pairs, each incompatible with the other, swap. The mathematics lets the swaps grow into cycles and then into chains, begun by a donor who gives to no one in particular.

The National Kidney Registry reported a chain that began in 2008 and produced thirty transplants across seventeen hospitals in eleven states over four months (Source Notes S58). The University of Alabama at Birmingham announced a long chain in which 74 donors and 74 recipients took part, later describing more than a hundred recipients (Source Notes S58). The Registry says that in 2024 it facilitated 1,744 transplants, had by then passed its ten-thousandth living-donor transplant, and was expected to account for roughly 30 percent of living-donor kidney transplants in the country in 2025 (Source Notes S59). Those figures come from the organization itself.

I mention the chain for the lesson, not the statistics. The chain is a tool that turned a refusal, the wrong blood type, into a transplant. It was an invention, it worked, and it scaled, and it is a good example of the orphan innovation escaping. The reason, I think, is that its benefit and its cost both landed on a party that could manage them: a nonprofit registry whose whole job was to make the matches.

The silence of asking

There is another barrier, and it costs nothing to fix.

Most people on the list never ask anyone for a kidney. The reasons are human. They do not want to be a burden, and they cannot imagine saying the words. A program developed at Johns Hopkins, called Live Donor Champion, was built around this silence: instead of asking the patient to ask, it trains a friend or relative to speak for her. In a pilot with fifteen candidates and fifteen matched controls, and a follow-up of 163 candidates in which 81 produced referrals of potential donors, the program was associated with roughly 5.5 times the odds of a donor referral (Source Notes S60). The evidence is from conference abstracts and registry entries, and there is no randomized trial that I found. But the program costs little, and a randomized trial would be easy to do.

A final barrier is time. One paper applying process-improvement methods to the living-donor evaluation reported that about a quarter of potential living-donor transplants are missed for reasons related to inadequate timing (Source

Notes S61). That is a finding with a familiar shape. The person is willing; the kidney is a match; and the process takes so long that the moment passes, the donor's circumstances change, or the recipient gets too sick.

What must be done

Finish the law. The living-donor reimbursement program created by the 2026 act should reimburse promptly, not after months, with an advance for travel and lodging; and the ceiling should be set against the real costs, with an annual adjustment, because a ceiling that one donor in five hits is a ceiling that is too low. The program saves money. It can afford to be generous.

Make a champion available to every candidate. Every transplant center should offer the champion model, or something like it, to every candidate, and a randomized trial of it should be run. It is the cheapest intervention in this book.

Shorten the evaluation. Each center should publish the time from a potential donor's first call to a decision, and set a target. The process-improvement literature shows it can be shortened.

Protect the donor. Job-protected leave, protection from insurance discrimination and life-insurance coverage for donors should be the baseline in every state, not a patchwork. The American Kidney Fund's report card is the right scoreboard. It should be a matter of federal law.

Study the tax credit honestly. A pilot of a donor credit, with an independent evaluation, would answer the questions that the argument has raised: does it increase donation, who benefits, and does it distort the gift? I would rather know than argue.

Teodoro, in the end, gives his brother a kidney. He loses six weeks of pay and, though he does not tell his brother, nearly his truck. What he gets is a brother at the next Thanksgiving. The system that gained fifty-five thousand dollars a year from his decision will never know his name. It is a small thing to ask that, the next time, the system say thank you and mean it.

CHAPTER 9

The Pump

> *"Hope deferred maketh the heart sick: but when the desire cometh, it is a tree of life."*
>
> Proverbs 13:12 (King James Version, 1611)

In this chapter. *Question:* what do we do with a tool whose benefit was proven sixteen years ago? *Answer:* we use it for fewer than four kidneys in ten, mostly where an organization that did not choose to bear the cost agrees to provide it. The pump is the clearest case in this book of an orphan innovation.

Here is the best experiment in the history of kidney preservation, and it begins with a coin.

In the years before 2009, a team of researchers in the Netherlands, led by Cyril Moers and working with transplant centers in six European countries, did something simple and beautiful. They waited for a deceased donor whose two kidneys would be transplanted into two different patients. Then they flipped a coin. One kidney went into a machine that kept cold preservation fluid flowing through it, called a hypothermic machine perfusion pump. The other was packed in ice, the way kidneys had been packed for fifty years. Same donor, same body, same night, same hospital system, same cold. Only the method of preservation differed. They did this 336 times, which means that 672 patients received kidneys in the trial (Source Notes S62).

The design was clever for a reason that is easy to miss. The great problem in comparing how kidneys are preserved is that kidneys differ. A kidney from a healthy twenty-five-year-old will do well in almost any condition; a kidney from a sixty-eight-year-old with high blood pressure will not. Compare pumped kidneys with iced kidneys across many donors, and the differences between the donors drown the difference between the methods. But two kidneys from the same donor are as close to identical as nature allows.

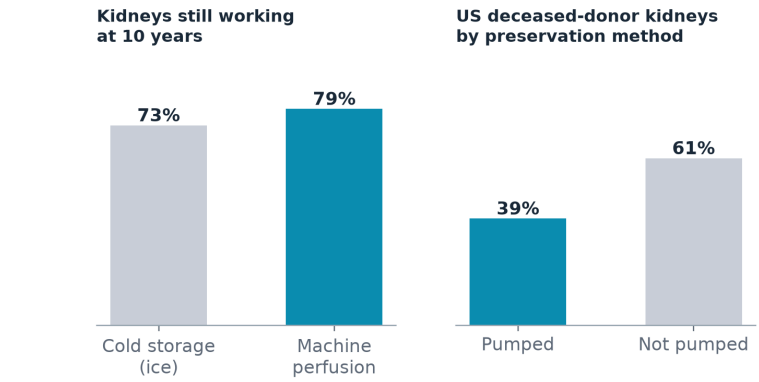
What the coin showed

The result, published in the *New England Journal of Medicine* in January 2009, was this: delayed graft function, the need for dialysis in the first week after transplantation, occurred in 70 of the pumped kidneys and in 89 of the iced ones, with an adjusted odds ratio of 0.57 (Source Notes S62). That is about 20.8 percent against 26.5 percent. The pumped kidneys also had better graft survival at one year.

The trial's first-year result made headlines in transplant circles and then did what such results do: it entered the literature and slowly altered practice.

Sixteen years later, in November 2025, the same group published a ten-year follow-up. It appeared as a short correspondence in the same journal and reported that the benefit had persisted: more kidneys that had been pumped were still working a decade later than those that had been iced. A press release from the investigators' university put the figures at 79 percent against 73 percent. The device's manufacturer has summarized the result as a 27 percent lower risk of graft failure over the ten years (Source Notes S63). Because the manufacturer is not neutral, I give that figure with a flag, and I note that the investigators' own figure shows a six-point gap.

There is an old principle in medicine that an intervention proven in a randomized trial should become the standard of care within a few years. The pump has a randomized trial, a decade of follow-up, the consistent testimony of surgeons, and a recent survey in which 92 percent of respondents endorsed it for helping them choose which kidneys to use and 90 percent endorsed it for reducing delayed graft function (Source Notes S64).



Left: 10-year follow-up of the Machine Preservation Trial (investigators' summary). Right: OPTN-based analysis, data through 2023.

Figure 9.1. Left: kidneys still working at ten years, machine perfusion versus cold storage. Right: the share of United States deceased-donor kidneys that are machine-perfused.

What the country did with it

Then, in 2023, an analysis of the national registry found that the pump was used for about 39 percent of deceased-donor kidneys (Source Notes S31).

Thirty-nine percent. A second analysis, a 2026 survey of eighty-eight transplant centers that use the pump, put the pattern in sharper relief. Centers that use it represent 35 percent of the country's active kidney programs and 63 percent of the volume. In other words, the busiest centers have it, and most of the small and mid-sized ones do not. And the pump is mostly not run by the centers. Most centers neither own pumps nor employ a dedicated perfusionist; about three-quarters of those surveyed said the pumping is done exclusively by the procurement organization (Source Notes S64).

The most-cited barrier in the survey, named by 58 percent of respondents, was the procurement organization *refusing* to pump. Not cost in the abstract. Not skepticism about the evidence. Refusal.

There is a second finding in that survey that I find more revealing than the first. Asked what they would do if a kidney they wanted pumped could not be pumped, 60 percent of the surgeons said they would *sometimes* decline the

kidney, and another 20 percent said they would. In other words, for four respondents in five, the absence of a pump is at least sometimes a reason to say no.

Connect that to the first chapters. The kidneys that are most likely to be discarded are the ones with the greatest uncertainty: older donors, biopsy findings, long cold times. Those are the kidneys for which a pump makes the most difference. Pumping both reduces the risk of delayed function and gives the surgeon a way to assess the kidney, since the resistance and flow measured during perfusion are informative. In the survey, 92 percent endorsed it for that purpose. A tool exists that would turn a marginal no into a confident yes, and its availability depends on whether an organization that is not the beneficiary chooses to provide it.

The orphan

This is what I mean by an orphan innovation.

Consider who gains from a pump. The recipient gains: fewer days on dialysis, a longer-lasting kidney. The transplant center gains: fewer complications, a more reliable result, better numbers on the scoreboard of Chapter 4. The Medicare program gains: a patient who leaves dialysis sooner. A kidney that would have been discarded and instead is transplanted gains the most.

Consider who pays. In the American system, as the survey indicates, it is frequently the procurement organization that bears the labor, the logistics and, I infer, much of the cost of running the machine. I could not find a published federal rule that says in plain terms who is responsible for paying for perfusion, or any published figure for what it costs the organization per kidney. The manufacturer claims a saving of nearly $10,000 per case from reduced delayed graft function, which is marketing and I treat it as such. What I can say is that the benefit of the pump accrues largely to someone other than the party that is asked to provide it.

That is the classic structure of a tool that does not spread. The person who would have to move it has no reason to; the person who would benefit has no way to. Its evidence is excellent. Its sponsor is no one.

A country that decided

In September 2022, Belgium made a different choice. Its national health insurer began reimbursing hypothermic machine perfusion for kidneys from older donors and from donors after circulatory death, on a national basis, through a national preservation service. A study published in 2025 analyzed the first year. Delayed graft function occurred in 14.4 percent of the pumped kidneys, which is well below the American rate. The national number of kidneys from donors after circulatory death that were transplanted rose from 90 to 175 a year, a change the authors called statistically significant; the number from older donors rose by less, and not significantly. The authors estimated annual savings of €3.59 million from reduced dialysis (Source Notes S65).

I need to hold this up to scrutiny. It was a retrospective analysis, not a trial; the numbers of kidneys are small; the savings estimate depends on assumptions I have not been able to read in detail; the vendor that supplies the service says it won the national tender. Belgium also differs from the United States in a thousand ways. But the contrast in structure is instructive. In Belgium, someone decided that the pump was a service, assigned an owner, set a price, and agreed to pay. In the United States, the pump is a favor.

Not every idea at the shelf deserves to win

I have a duty to say what has *not* worked, because a book that praises one innovation and ignores the failures of its neighbors is advocacy of the worst kind.

Warming the kidney while it is pumped, rather than cooling it, seemed a logical next step. A randomized trial of warm perfusion for kidneys from donors after circulatory death, published in 2023, found that delayed graft function occurred in 60.7 percent of the warm-perfused kidneys and 58.5 percent of the iced ones, no difference (Source Notes S66). A 2026 trial adding two hours of warm perfusion to the cold pump found no improvement in immediate function. A trial of adding oxygen to the cold pump for kidneys from older donors found no benefit, though it was underpowered. A 2025 review concluded that the effect of oxygen during perfusion has been mixed and the clinical benefit of warm perfusion unproven (Source Notes S66).

That is why the cold pump stands alone as proven and why the other tools in this chapter's neighborhood, including the oxygen carriers in the next, deserve to be examined with suspicion. The point of this book is not that every new tool is wonderful. It is that for one of them the evidence is in, and the system has not acted on it.

What must be done

Set a standard. The national network's kidney committee should define, within twelve months, which kidneys should be machine-perfused, by donor type, age and expected cold time, and require that perfusion be made available when a center requests it for a kidney in that group. The 2009 trial is old enough to have earned a standard.

Assign an owner and a price. The federal payer should state explicitly who pays for perfusion, and pay it. The Belgian model, a national service with a national reimbursement, is one template. A pass-through of the full cost to the organization that does the pumping is another. What cannot continue is an arrangement in which the cost falls on one party and the benefit on another.

Fix the thresholds. The survey found that acceptable resistance and flow cut-offs vary widely, and that inaccurate parameters were a recognized cause of kidneys being underused. The network should publish evidence-based thresholds, update them annually, and audit how they are used.

End the refusal. No procurement organization should be permitted to decline to pump a kidney a transplant center has requested for a recognized indication, except for a documented clinical or logistical reason, reported publicly.

Publish pump use. The percentage of each organization's and each center's kidneys that were perfused, and the outcomes, should appear in the dashboard of Chapter 3.

The pump has waited sixteen years. In a ten-year follow-up the people who waited with it lived with the difference. Next, we look at a tool that has not yet earned the same evidence, and has been waiting just as long.

CHAPTER 10

The Worm, the Cow, and the Blood That Wasn't

> *"But ask now the beasts, and they shall teach thee; and the fowls of the air, and they shall tell thee."*
>
> Job 12:7 (King James Version, 1611)

In this chapter. *Question:* can a molecule that carries oxygen outside the body help a kidney survive the cold, and why has the idea taken so long? *Answer:* it might, and no one yet knows. The first generation of such molecules failed spectacularly as a blood substitute, and the companies that made them mostly failed too. A second use, inside the organ's preservation fluid, avoids the old safety problem, and the evidence for it is early, small, and partly company-sourced. This chapter says exactly how far it goes.

By his own account, Franck Zal was looking out of a laboratory window at a beach.

It was at the marine station in Roscoff, on the coast of Brittany, where the tide goes out a long way. He was a biologist who studied the blood of sea creatures, and on the sand below him were thousands of lugworms, the fat, pink, burrowing animals that fishermen dig for bait. At low tide, when the water drained away and the sand began to dry, the worms lay in their burrows with almost no oxygen. They did not die. He asked himself how (Source Notes S67).

The answer, as he came to tell it, was in their blood. The lugworm carries its hemoglobin not inside cells, as humans do, but loose in the plasma, in giant molecules, much larger than ours. A human hemoglobin molecule can hold four

molecules of oxygen. By one description, a lugworm's can hold 156 (Source Notes S67). He made the case that such a molecule might help keep human tissue alive when blood could not reach it, and in 2007, with the backing of his research institutions, he founded a company, Hemarina, to find out.

I begin with Zal because the story is a good one and because the rest of this chapter will be less romantic. A good story is not evidence. It is the reason someone looked.

The cow, and the graveyard

To see why a molecule from a worm is a novelty and a molecule from a cow is a cautionary tale, you have to know the graveyard.

The idea of a blood substitute is old and appealing. Blood is scarce, perishable and has to be typed and matched. A solution of hemoglobin, the oxygen-carrying protein stripped out of red cells, would in theory need no matching and could be kept on a shelf. Researchers were infusing such solutions into animals more than a century ago. The fear of contaminated blood in the 1980s turned the idea from a curiosity into an industry (Source Notes S68).

The industry's history is the history of a handful of companies and one conclusion. Baxter, a major pharmaceutical company, developed a human hemoglobin chemically cross-linked into a stable form, and in a trial in trauma patients the product produced, by one secondary review's account, a 72 percent increase in deaths compared with saline. Hemosol, in Canada, developed a hemoglobin linked with a sugar-derived reagent; its late-stage trials were stopped in 2003 after cardiac events and the company went insolvent within a few years. Northfield Laboratories, in Illinois, developed a polymerized human hemoglobin and ran the field's boldest trial, in ambulances, in patients who could not consent; the regulatory outcome was unfavorable and the company filed for bankruptcy in 2009. Biopure, in Massachusetts, made two bovine products: one approved for dogs in the late 1990s, one for people in South Africa in 2001. In 2003 the American regulator halted a proposed trauma study and rejected an application for use in orthopedic surgery. The Securities and Exchange

Commission later found misleading statements about the regulator's stance, and in 2005 brought charges against the company's chief executive and others. On July 16, 2009, Biopure filed for bankruptcy protection (Source Notes S68).

The verdict on the whole class came in 2008, in a pooled analysis in the *Journal of the American Medical Association* by Charles Natanson and colleagues. It combined sixteen trials of five products in 3,711 patients. The treated patients died more often than the controls, 164 against 123, a relative risk of 1.30. They had more heart attacks, 59 against 16, a relative risk of 2.71 (Source Notes S69). The paper did not distinguish among molecules, and it did not need to. Regulators, editors and investors drew the obvious conclusion. The field went quiet.

I tell the story with some detail because the lesson of the graveyard is often misread. The lesson is not that oxygen carriers do not work. It is that injecting free hemoglobin into a person in shock appears to be dangerous, probably because the molecule scavenges the body's own nitric oxide, tightening blood vessels and stressing the heart. That is a lesson about a *route*, not about a molecule.

The dog

One product survived, in a way that tells you something. Oxyglobin, made from bovine hemoglobin, was approved in the United States for anemic dogs in the late 1990s, before any comparable product cleared a regulator for humans anywhere in the wealthy world. It sat in a bag on a shelf, needed no cross-match, and could be hung by a tired clinician at two in the morning. It also, in time, disappeared from clinics when its manufacturer's finances failed, leaving veterinarians to search the country for the last units.

The dog got there first because the question asked of the dog was easier. A dog in crisis with no donor blood has an alternative of nothing, and a regulator can approve on the sign of improvement. A human regulator must be convinced that the product beats red cells. The difference between the two questions is the difference between the two histories (Source Notes S68). I mention it because it points to the thing the molecule can do, and the thing the market could not support.

A different route

Now consider a different place to put the molecule.

A kidney on a pump, or in a cooler, is outside the body. Its cells are starving, as we saw in Chapter 3, and the damage is done by lack of oxygen and by what happens when oxygen returns. Suppose the preservation fluid in which the kidney sits, or through which it is pumped, contains an oxygen carrier. The molecule is not injected into a patient in shock. It is in the kidney's bath. The exposure of the patient who receives the organ is a fraction of what was tested in the trauma trials. The risk is not zero. It is a different, much smaller regulatory and safety question.

That is the design behind the product from Hemarina, called HEMO2life, built on a purified form of the worm's hemoglobin, called M101. In a study called OXYOP, which ran at French transplant centers from 2016, sixty kidneys preserved with the additive were compared with the matching kidney from the same donor preserved in the usual way (Source Notes S70). The investigators' follow-up reported that patient survival at four years was 98.3 percent in the group that received the additive, against 86 percent among controls, a difference with a p-value of 0.016. The manufacturer reports less delayed graft function in some subgroups. The product received a European marketing mark, and has been used in other settings, including a 2021 double upper-limb transplant in India (Source Notes S70).

I cannot say how seriously to take these numbers, and I want to tell you why.

First, the study was small. Second, the paired design, which is a strength of the Dutch pump trial, was in this case compromised: by the way the study was arranged, the cold time was longer in the control kidneys than in the treated ones, and cold time alone changes outcomes (Source Notes S70). A benefit that might be partly due to a shorter cold time is not a clean benefit. Third, the survival figure is a four-year follow-up of recipients, an outcome that depends on many things other than the kidney. Fourth, the sources are mostly the company and its distributors, and a published paper from the group discloses that its founder has a financial interest. Fifth, the sources disagree about how many patients took part, sixty or one hundred and twenty.

A randomized trial, OxyOp2, was designed to answer the question properly, and its protocol was published in 2023. I could find no results (Source Notes S70).

None of that makes the work worthless. It makes it a first signal, the kind that deserves a larger trial.

The program

A second effort is closer to home. A company called BHOC Therapeutics is developing cell-free bovine hemoglobin carriers, and its transplant program, BHOC Transplant, describes a research plan: whether such a carrier could keep a donated organ supplied with oxygen through preservation, machine perfusion and the return to body temperature, beginning with one organ, a kidney, and one protocol, with every measurement defined in advance. The Foundation I founded has written about it, and I describe it here as its sponsors describe it: as a research concept. They state that it is not an approved treatment, that BHOC has not been shown to prevent rejection or change transplant outcomes, and that results and time gained are still to be proven (Source Notes S71).

I like that language, and I would like it applied to every other claim in this chapter. A claim of benefit without a trial is a hypothesis. The most valuable thing a patient community can do with a hypothesis is insist on the experiment.

What the autopsies teach

The five failed companies teach something beyond their molecules, and a few of the lessons apply to anyone watching this field today.

The *single-product company* has nothing to absorb a bad trial; every one of these companies was built on one molecule. *Judgment by class* means a failure of one product taints all of them: a bovine polymer is not chemically a cross-linked human protein, but the 2008 analysis did not distinguish. *Governance failures* are scientific variables: a sponsor caught shading its statements pays a tax on every later claim. *Improving comparators* mean the blood supply became safer, shrinking the original advantage. And *survival through niches* suggests that a modest market,

veterinary use, a few foreign approvals, compassionate access, can keep alive a manufacturing process, a regulatory file and a clinical record that no start-up could build from nothing (Source Notes S68).

There is also a simple checklist, which an outsider can apply from public documents. Watch the ratio of announcements to publications. Watch how the company speaks about the regulator. Watch whether the first trial is an all-or-nothing wager. Watch whether the company can describe its manufacturing plainly. Watch the cash runway against the date the trial will read out.

What must be done

Run the trial. A properly designed, independently funded, registered randomized trial of an oxygen carrier in kidney preservation is overdue. It should use paired kidneys with equalized cold time, preregistered primary outcomes, delayed graft function and one-year kidney function with graft survival at five years, blinded adjudication, and publication of results whatever they are. By my arithmetic, a trial of unpaired recipients, hoping to cut delayed graft function from 26 percent to 18 percent, would need several hundred patients per arm; a paired design needs fewer. That is a modest sum by the standards of drug development, and well within the reach of a public funder.

Pay for evidence, not for hope. A federal payer can offer coverage on condition that patients are enrolled in a registry or a trial, so that use produces knowledge.

Protect the molecule from the company. The oxygen-carrier field has repeatedly lost working science because a sponsor failed. Public funders, including the defense and health agencies with an interest in blood substitutes, should hold the dossiers, manufacturing know-how and clinical records of products that show promise, so that the next bankruptcy does not erase them.

Seat the patients. Every trial should include patients and donor families in design and oversight. A trial that patients trust enrolls faster and drops out less.

The worm and the cow are, for now, a possibility. The next chapter turns to two larger possibilities, a gene-edited pig kidney and a cell therapy that could end the need for anti-rejection drugs.

CHAPTER 11

The Pig, the Cell, and the Waiting Room

"In the fields of observation chance favors only the prepared mind."

Louis Pasteur, lecture at the University of Lille (1854)

> **In this chapter.** *Question:* if gene-edited pig kidneys and cell therapies that end the need for anti-rejection drugs both arrive, does the argument of this book become obsolete? *Answer:* no. It becomes urgent. Each technology would remove one old constraint, the clock or the lifelong drugs, and each would walk straight into the structure of Chapters 4 through 7. The question is whether the structure will be ready.

Count to 271.

That is how many days a man named Tim Andrews lived without dialysis on a kidney taken from a pig. The operation was done at Massachusetts General Hospital in January of 2025. The organ came from an animal whose genome had been edited by the company eGenesis so that it carried human genes and lacked several of its own, in the hope that a human immune system would leave it alone. It worked for 271 days, a record for such a kidney, until it began to fail and was removed in October (Source Notes S72). In January of 2026, according to Harvard Medical School's account, Andrews received a human kidney, becoming the first person to go from a pig kidney to a human one (Source Notes S72).

Let that number sit for a moment. For more than eight months a man who had been sentenced to a chair three afternoons a week was free of it. He could plan a week with no machine in it. When the pig kidney failed, he did not die. He went back to dialysis, and a few months later he was given a human organ. In the history of this disease there was, until recently, nothing like that.

I put this chapter in the middle of the book on purpose. The first half described a system that wastes the gifts it has. The second half describes the tools that might change it. Before I turn to what must be done, I owe you an honest look at the two most dramatic tools on the horizon, because a reader who has come this far will reasonably ask whether any of this matters if the shortage is about to be solved by a farm.

What has actually happened

The record is short, and I will keep to what I can support. In March of 2024 surgeons at Massachusetts General transplanted a gene-edited pig kidney into a man named Richard Slayman, who died two months later; the hospital said at the time there was no indication that his death was the result of the transplant. A second patient, Towana Looney, received a pig kidney at NYU Langone in November 2024 and lived with it for roughly four months before it was removed because of rejection (Source Notes S73). These accounts come from news coverage, and I could not reconfirm every date in the sources open to me.

In the autumn of 2025 the regulators moved. The Food and Drug Administration cleared formal clinical trials in living patients. eGenesis received approval for a trial in September, and in November United Therapeutics announced the first transplant in its own trial, called EXPAND, of a kidney with ten gene edits, six human genes added and four pig genes knocked out, performed at NYU Langone, with twenty-four weeks of intensive monitoring and lifelong follow-up (Source Notes S74). A report in *Nature Medicine* in 2025 profiled the immune response in a living recipient of a gene-edited pig kidney, and a 2026 review in the *American Journal of Transplantation* surveyed the field (Source Notes S72).

That is the state of knowledge as of this writing: a handful of patients, a handful of months, a formal trial just begun. Nobody can say how it ends. But the direction of travel is clear enough to take seriously, and the first question to ask of any technology is what it would change.

What a farm would change

The first thing is the clock. A pig kidney is not a gift that arrives at three in the morning from a stranger who has just died. It is a product, raised under controlled conditions, available on a schedule. Proponents argue that this means scheduled surgery, a recipient prepared and well, and, as a direct consequence, almost no cold ischemia time: the kidney comes out of the animal and goes into the patient in a controlled sequence. If that holds, then a large part of Chapter 3, the clock with no owner, would dissolve.

The second is supply. The list is not short of patients. It is short of organs, and a farm can in principle raise more.

The third is the discard. A kidney that is raised to a specification does not have a biopsy at two in the morning, an uncertain donor history, or a 58-year-old donor with some scarring. The factors that produce the 29.3 percent nonuse rate would, for this class of kidney, mostly not exist.

I find these genuinely hopeful. I do not think anyone should dismiss them, and I do not think anyone should be sure of them.

What a farm would not change

Now the sobering part.

A pig kidney would still be offered to a patient, not given. It would still have to be accepted or declined by a surgeon, who would still have a scoreboard. Whether a xenotransplant counts in a program's published results, in the risk-adjusted outcomes that fed the pressure described in Chapter 4, is a question I could not find answered in any source. It should be answered *before* the first commercial transplant, not after the first program is flagged. If a failed pig-kidney

transplant counts against a center the way a failed human one does, the invisible ledger will treat the pig exactly as it treated the marginal human kidney, and the surgeons who most want to use it will have the least reason to.

It would still have to be paid for. I found no published price. A technology that costs more than the dialysis it replaces will meet the five tests of Chapter 12 at a disadvantage; one that costs less will meet them with a tailwind. And it would still reach patients through the same hospitals and the same referral chains, which means the people who already have the best access to transplantation would be first in line. Chapter 2's lesson, that a kidney is lost in a queue before it is lost in a cooler, applies to a pig as to a person.

And it would still, probably, require immunosuppression. This is where the second technology enters.

The cell

The reason transplant recipients take drugs every day for life is that the immune system sees a transplanted kidney as foreign and tries to destroy it. Those drugs work, and they cost. They increase the risk of infection and cancer, they have side effects of their own, and a patient who cannot afford them can lose the kidney. The graft survival figures from the registry, a median of about 11.7 years for kidneys from deceased donors and about 19.2 for living donors (Source Notes S52), look good, and they are also a measure of how long a body will tolerate a foreign organ under chemical restraint.

The dream of the field for fifty years has been *tolerance*: teaching the body to accept the organ as its own, so that the drugs can be stopped. The progress is real and recent. A phase 3 trial of a cell therapy called MDR-101, developed by Medeor Therapeutics for recipients of kidneys from certain living donors, treated 20 patients and compared them with 10 controls. In 2023 an interim analysis reported that 16 of 19 treated patients had been able to stop all immunosuppression at two years. The final results, published in the *American Journal of Transplantation* in July 2025, reported that about 95 percent of treated patients had been able to stop all immunosuppression roughly a year after treatment, and about 75 percent remained off the drugs for more than two years;

four had to resume them (Source Notes S75). A nephrologist at a leading center, interviewed in April 2026, described tolerance as the frontier of kidney care (Source Notes S76).

The numbers are small. The therapy applies to a narrow group. The control group is ten patients. I would not tell a reader that the problem is solved. But think about what it would mean: a person who has taken a handful of pills every day for twenty years, who has been through the infections and the cancers and the monthly bill, could someday walk away from all of it. That is a different kind of freedom from the one a pig kidney offers, and in some ways a deeper one.

The waiting room

Both technologies share a waiting room, and it is the one this book has been describing.

Whoever manages the first fifty xenotransplants will decide something important by accident: who the technology is for. The trials have inclusion criteria, and the people who meet them are not a random sample of the waiting list. Whoever receives the first cell therapy will be those with a living donor, which means those with a family willing and able to give. Neither is a scandal. Both will have consequences for fairness, and consequences that are easy to see now and hard to reverse later.

This is why I ask, in the fourth part of this book, that trials of new technologies report their enrollment by race, geography, income and sex, as they go. A technology that reaches only the well connected will deepen the divide it was meant to heal.

What must be done

Decide the scoreboard first. The federal agencies that oversee transplant programs should state, before the first commercial xenotransplant, how it will count in program evaluations, and should protect centers that adopt it responsibly from penalty for expected early failures.

Publish the enrollment. Every xenotransplantation and tolerance trial should publicly report who is enrolled, by race, region, income and sex, quarterly.

Plan the payment. The federal payer should publish, in advance, how it will pay for a xenotransplant and a tolerance therapy, so that the first patients are not chosen by their insurance.

Prepare the system, not just the procedure. The pump, the offer ledger and the fifteen moves of Part Four matter even if every promise of this chapter comes true, because the structure that a miracle will enter is the one we have, and it should be a good one.

Pasteur was talking about scientists. I think the sentence applies to systems. Chance, in the form of a pig, a cell or a trial, will favor only the system that is ready.

The last chapter of this part steps back from individual tools and asks why some good ideas spread in months and others take a generation, and how we can tell which of the slow ones deserve to win.

CHAPTER 12

The Fast Idea and the Slow Idea

"The reformer has enemies in all who profit by the old order, and only lukewarm defenders in all those who would profit by the new order."

Niccolò Machiavelli, The Prince, chapter 6 (Thomson translation, 1910)

In this chapter. *Question:* why do some good ideas spread in months and others in generations, and how can we tell a slow idea that deserves to win from one that deserves to lose? *Answer:* speed depends less on the quality of the evidence than on five practical tests, and the same tests that explain why the pump waited also explain why a cheap, appealing idea about dialysis temperature was adopted by intuition and then failed a large trial.

In October of 1846, in the amphitheater of a Boston hospital, a dentist named William Morton put a man to sleep with ether while a surgeon removed a tumor from his neck. The patient felt, he said afterward, nothing. The news crossed the Atlantic within weeks. Within months, surgeons in European capitals were using ether. Within a few years, nearly every operating room in the Western world was (Source Notes S77).

Seven months after that, in the spring of 1847, in a maternity hospital in Vienna, a young obstetrician named Ignaz Semmelweis noticed that women delivered by medical students died of childbed fever at a far higher rate than women delivered by midwives, and he made a guess about why. The students came to the delivery ward straight from the autopsy room. He ordered them to wash their hands in a solution of chlorinated lime. By the commonly cited figures, the death rate in his ward fell in two months from about 18 percent to about 2

percent. His colleagues largely refused to believe him. He lost his post, and died in 1865 in an asylum. The idea that germs cause infection, and that surgeons should clean their hands and instruments, took most of a generation to become routine (Source Notes S77).

Same century, same hospitals, same profession, two ideas. One spread in months. The other took decades.

Why anesthesia won

The surgeon and writer Atul Gawande asked this question in an essay called "Slow Ideas" in *The New Yorker* in 2013, and his answer is the best short theory of adoption I know. Anesthesia, he argued, was an obvious, immediate, visible benefit to everyone in the room. The patient stopped screaming. The surgeon could work slowly. The benefit arrived the moment the idea was used and could be seen by anyone watching. Antisepsis, by contrast, demanded effort now for a benefit later, invisible and statistical: "pain now for gain later," in the phrase quoted from him. A surgeon had to fuss with carbolic acid and scrub, and the reward was an infection that did not happen, which no one could see. It took regulation and a change of professional culture to make the practice stick (Source Notes S77).

Gawande's contrast has a corollary that matters here. A review of the essay notes that money does not explain the difference: the incentives ran in the right direction for both ideas. What mattered was the shape of the benefit: who could see it, how fast, and what it cost the person asked to change.

The surgeon with a marginal kidney, the organization with a pump, the dialysis manager with a new machine, are all in the position of the nineteenth-century surgeon deciding whether to wash his hands.

Five tests

From this, and from the cases in the earlier chapters, I have built a short method. I use it as a checklist and not as a law. Ask five questions of any new tool.

One: is the evidence good? Randomized, replicated, with meaningful follow-up, not a company's press release.

Two: can the person who must act see the benefit? A pump that prevents a complication a surgeon will see in a patient eight days later passes. A tool whose benefit shows up as a statistical improvement in outcomes five years from now fails.

Three: does the person who bears the cost also gain? If the benefit goes to one party and the cost to another, the idea is an orphan.

Four: does it improve a number someone is judged on? A tool that raises a program's scoreboard result will be adopted. A tool that improves lives but not a metric will not.

Five: does it require someone to change a routine at 3 a.m.? Every tool that must work in an operating room at night, or in a courier chain, or in a dialysis unit's third shift, has to work for the person who is tired.

Now apply them.

Idea	Evidence	Benefit visible to actor?	Payer also gains?	Moves a metric?	Easy at 3 a.m.?	Speed
Hypothermic pump	Strong: randomized trial, 10-year follow-up	Partly	No: procurement organization pays	Partly	No	Slow
Hemodiafiltration	Strong: large randomized trial, one	Not to the clinic	No: fixed payment per treatment	No	Moderate	Slow
Kidneys from hepatitis C donors	Moderate: small trials, then registries	Yes: more transplants	Yes	Partly	Moderate	Moderate
Oxygen carrier additive	Early, partly company-sourced	Unknown	Unknown	Unknown	Unknown	Not yet testable
Offer filters	Pilot data	Yes: fewer calls	Yes	Yes	Yes	Slow, then approved
Cooler dialysis fluid	Weak, then strongly negative	Yes: patients feel better	Yes	No	Yes	Fast, then wrong

The table is crude, and the entries in several cells are my judgment. But look at the pattern. The pump, the best-evidenced idea of the lot, fails three of the five tests. The idea that was easiest to adopt, and was adopted by intuition, is the one whose largest trial found nothing.

The idea that deserved to lose

For years, nephrologists had observed that some patients felt better, with fewer drops in blood pressure, when the fluid used in dialysis was a little cooler than body temperature. Small studies supported the idea, some with dozens of patients. The reasoning was sensible: cooler blood constricts vessels and holds pressure. The change was almost free. A unit could implement it in an afternoon, and patients who felt better said so.

In 2022 the largest trial ever done in hemodialysis tested it. Called MyTEMP, it was a cluster-randomized trial across dialysis centers in Ontario, using the province's health records, and it compared a personalized cooler dialysate, 0.5 to 0.9 degrees below each patient's own temperature, with a standard fluid at 36.5 degrees. The outcome was cardiovascular death or hospital admission for a heart attack, stroke or heart failure. It occurred in 21.4 percent of patients receiving the cooler fluid and 22.4 percent of those receiving the standard, a difference too small to be distinguished from chance. Blood-pressure drops during treatment were the same (Source Notes S4). The authors concluded that there was no justification for adopting cooler fluid for all patients, and noted that some patients found it uncomfortable.

John Stuart Mill wrote that "he who knows only his own side of the case, knows little of that." I have told you much of one side of the case in this book, the side that says the system is too slow. MyTEMP is the other side. A system that adopts every plausible idea at once would be as bad as one that adopts none, and for the same reason: it has no way of finding out.

The lesson is not that cooler dialysis is useless. Some patients may benefit, and the trial measured heart events, not how a patient feels. The lesson is that intuition is a poor guide, and that fast adoption without a trial is just a different kind of failure.

Why evidence alone does not win

The physicist Max Planck, in the last pages of his autobiography, offered a bleak account of how scientific ideas succeed. "A new scientific truth does not triumph by convincing its opponents and making them see the light," he wrote, "but

rather because its opponents eventually die, and a new generation grows up that is familiar with it."

It is a funny sentence, and the field of kidney medicine cannot afford it. Planck's remedy, waiting for a generation to pass, is a remedy for physicists. Patients on the list die on a shorter schedule. In 2024 alone, 3,743 candidates were removed from the kidney waiting list because they died (Source Notes S6).

What the five tests suggest is that adoption is not mostly a matter of persuasion. It is a matter of design. A tool becomes routine when the person who must use it can see the benefit, pays a fair price, is judged by the result, and finds it easy to do at night. None of these requires waiting for anyone to die. All of them require someone to decide that adoption is their job.

The owner of the adoption

And that is the missing piece. In the case of the pump, the oxygen carrier, hemodiafiltration, and the offer ledger of Chapter 2, no institution has the task of moving a proven idea from the journal to the bedside. The journals publish, the regulators approve, the payers reimburse, the societies recommend. None of them owns the gap between what is known and what is done.

What must be done

Make adoption someone's job. For any intervention with randomized evidence of benefit in kidney disease or transplantation, the responsible federal agency should be required to publish, within twelve months, an adoption plan: who pays, who owns the process, who measures, and by what date. If it chooses not to adopt, it should say why.

Run the cheap, large trial. MyTEMP was possible because Ontario's dialysis centers were part of a registry that could randomize at the level of the center and measure outcomes through health records. The United States has the data infrastructure to do the same. A national program of pragmatic trials of plausible dialysis practices, including temperature, session length, hemodiafiltration and home therapy, would turn intuition into evidence quickly and cheaply.

Apply the five tests in public. Whenever a federal agency or professional society reviews a new tool, it should publish how the tool scores on the five tests, so that the barriers to adoption are visible and can be addressed.

Sunset the ideas that fail. A practice that has been tested and found useless should be removed from guidelines and payment, and the people who adopted it should hear about it.

Slow ideas and fast ideas are not opposites. They are two ways of ignoring the evidence. The fourth part of this book is about the decisions that follow from taking it seriously.

PART FOUR

THE WAY OUT

From Anger to Plan

"As our case is new, so we must think anew, and act anew."

Abraham Lincoln, Annual Message to Congress (1862)

Most books of this kind end with an appeal to awareness. I have little patience for it, and the reason is that the people who will read this book are, for the most part, already aware. They have waited, or watched someone wait. What they lack is not information about the problem but a short list of specific, answerable demands, each addressed to a named institution, with a number attached and a date beside it.

Part Four is that list. Chapter 13 sets out fifteen moves, grouped by the institution that must make them. For each there is a mechanism, the evidence from the earlier chapters that supports it, a risk, and a test you can check in a year. Chapter 14 turns the list into a ninety-day plan for a patient, a care partner, a clinic or a community group, with twelve questions to ask, three letters to send, and a one-page log to keep. Chapter 15 steps back one more time and asks what we owe a gift.

A word about anger. It is a legitimate response to a kidney that never reached a body and a person who never left the chair. But anger without a plan is a feeling. With a plan, it is fuel.

CHAPTER 13

Fifteen Moves

"Power concedes nothing without a demand. It never did and it never will."

Frederick Douglass, speech on West India Emancipation (1857)

> **In this chapter.** *Question:* what, exactly, should be done, by whom, and how will we know it worked? *Answer:* fifteen moves in five groups, each with an owner, a mechanism, the evidence behind it, and a test you can check in a year. Nothing here overrules a surgeon's judgment about a patient. Everything here changes the structure that surrounds the judgment.

A demand has three parts: a request, an addressee and a date. Most advocacy fails because it has one of the three. This chapter tries to supply all of them.

Before the list, a word about what I am not proposing. I am not proposing that surgeons be forced to accept kidneys they believe will harm a patient. I am not proposing to eliminate the one-year outcome measure or to abandon oversight of procurement. I am not proposing any technology as a cure. Every move below is a change to payment, information, accountability or evidence, the four levers that, in the earlier chapters, appear to govern behavior.

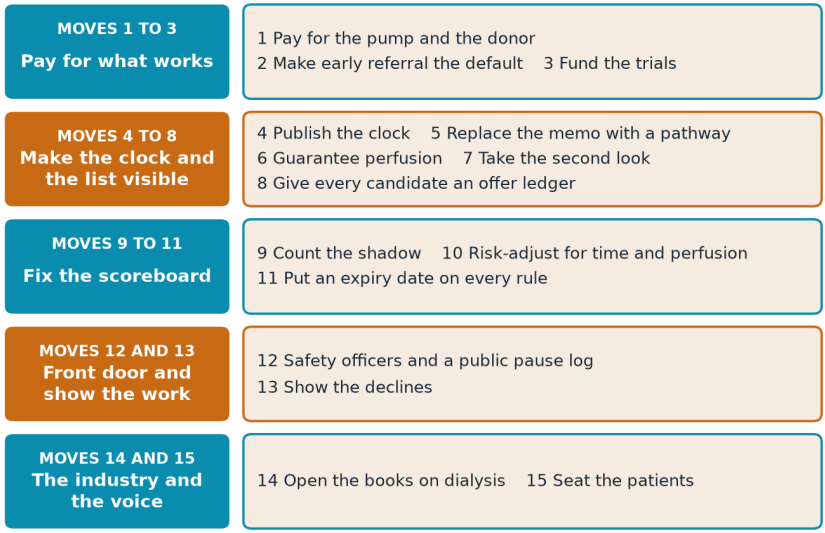


Figure 13.1. The fifteen moves, grouped by who must act, with the chapter in which each is argued.

#	Move	Who must act	First test, within 12 months
1	Pay for the pump and the donor	CMS, HRSA, Congress	A published rule stating who pays; donor reimbursement paid promptly
2	Make early referral the default	CMS, dialysis facilities	Referral within 90 days of starting dialysis, reported by facility
3	Fund the trials	NIH, CMS, defense agencies	Three registered pragmatic trials launched
4	Publish the clock	HRSA, OPTN	Public monthly cold-time dashboard
5	Replace the memo with a pathway	HRSA, OPTN	Expedited placement route with public audit
6	Standardize and guarantee perfusion	OPTN	National thresholds; refusal reported
7	Take the second look	OPTN	Automatic rerouting after a set number of declines
8	Give every candidate an offer ledger	OPTN, centers	First quarterly statements sent

#	Move	Who must act	First test, within 12 months
9	Count the shadow	SRTR, CMS	Offer acceptance ratio includes discarded organs
10	Risk-adjust for time and perfusion	SRTR	Adjusted model published
11	Put an expiry date on every rule	HRSA, CMS	Sunset clauses in all new metrics
12	Protect the front door	OPOs, HRSA	Safety officers and a public pause log
13	Show the declines	Centers	Decline reasons and downstream outcomes public
14	Open the books on dialysis	CMS, FTC, Congress	Ownership and compensation database; market review
15	Seat the patients	Everyone	Patients and donor families on every board and trial

Group one: pay for what works (moves 1 to 3)

1. Pay for the pump and the donor. In two places in this book the benefit of a proven thing lands on one party and its cost on another. The pump's cost falls on a procurement organization that does not gain; the living donor's cost falls on a person whose gift saves the payer about $55,000 a year (Chapter 8). The federal payer should state in a published rule who pays for kidney perfusion and pay it, either through a pass-through to procurement organizations or a payment to centers, and the living-donor reimbursement created in 2026 should pay promptly, in advance for travel and lodging, with a ceiling set against real costs (Source Notes S55). Belgium's model, a national service with a national reimbursement, produced a rise in transplanted donation-after-circulatory-death kidneys from 90 to 175 a year (Source Notes S65). *Evidence:* Chapter 9. *Risk:* cost; mitigated by the savings from reduced delayed graft function, which one Belgian estimate put at €3.59 million a year. *Test:* a rule published, and the share of kidneys perfused rising from 39 percent toward a published target.

2. Make early referral the default. A trial across 134 Georgia dialysis facilities and more than 9,000 patients found that a structured referral intervention improved referral for evaluation and narrowed the racial gap (Source Notes S51). That should be the standard of care: every patient informed, and referred for evaluation within ninety days of starting dialysis unless there is a documented reason, with the percentage published by facility. *Risk:* referral is not transplant, and a flood of referrals could lengthen evaluation queues. *Test:* the share of new dialysis patients referred within ninety days, by facility.

3. Fund the trials. For the oxygen carriers of Chapter 10, hemodiafiltration, home dialysis, session length and other plausible tools, public money should buy the experiment that no sponsor will. Coverage with evidence development, in which a payer covers a new tool on condition that patients are enrolled in a registry or a trial, can turn use into knowledge. *Evidence:* MyTEMP shows that a very large, pragmatic trial in hemodialysis is feasible at the level of the facility using health records (Source Notes S4). *Test:* three registered trials launched with public funding.

Group two: make the clock and the list visible (moves 4 to 8)

4. Publish the clock. Cold time per kidney, per procurement organization, per center and per route, updated monthly, with a named organ custodian responsible from recovery to use. The 2021 allocation change added about 2.9 hours of cold time to regional sharing (Source Notes S21). A public goal of recovering those hours would turn a complaint into an engineering target. *Test:* a public dashboard and a median cold time that moves.

5. Replace the memo with a pathway. Define an expedited placement route for kidneys that have been declined a set number of times, with criteria, real-time logging and a public audit, so that fairness to waiting candidates and use of hard-to-place kidneys are served together. The network's working group has already drafted a definition of an offer and criteria for an expedited pathway (Source Notes S34). It should be finished, approved and tested. And any federal communication likely to change allocation behavior should carry a forecast of its effect on transplants and a thirty-day review. *Evidence:* Chapter 5. *Test:* a pathway in place and a first public audit.

6. Standardize and guarantee perfusion. Evidence-based thresholds for pump resistance and flow, published and updated annually; a rule that no procurement organization may refuse a transplant center's request to pump a kidney in a recognized indication without a documented reason, reported publicly. In the survey of centers, refusal was the most-cited barrier at 58 percent (Source Notes S64). *Test:* thresholds published; refusals counted.

7. Take the second look. After a kidney has been declined a set number of times, automatically route it to the programs with the best records of using hard-to-place kidneys. The offer-filter pilot showed that centers could avoid two-thirds of the offers they would otherwise receive and, in some cases, increase their transplant volumes (Source Notes S18). Routing is the other side of the same coin: sending fewer offers to centers that never accept them, and more to centers that do. *Test:* the median number of offers before acceptance, which has been seven (Source Notes S14), falling.

8. Give every candidate an offer ledger. A quarterly statement to each person on the list showing how many kidneys were offered on her behalf, how many were declined, the reasons, and what became of them. It would not oblige any center to accept an offer. It would oblige the system to let the person whose life depends on the decisions see them. The research on 14 million offers found that candidates who died had received a median of sixteen (Source Notes S1). Most of those candidates never knew. *Risk:* confusion and anxiety; mitigated by a plain-language explanation and a coordinator available to discuss it. *Test:* first statements delivered, and the share of candidates who say they understood them.

Group three: fix the scoreboard (moves 9 to 11)

9. Count the shadow. The offer acceptance ratio, which now drives federal review and a portion of payment, excludes offers for kidneys that were eventually discarded (Source Notes S27). It should include them. The registry should also publish, for each program, the number of kidneys it declined that were transplanted elsewhere with good results. *Test:* a revised ratio.

10. Risk-adjust for time and perfusion. The formula that predicts a program's expected results should account for cold time, delayed graft function and whether a kidney was machine-perfused, so that a program that uses a hard kidney well is credited for the risk. A published safe-harbor rule would let programs accept a defined class of difficult kidneys without being penalized for the expected share that fail. *Test:* the adjusted model published and its effect on program rankings reported.

11. Put an expiry date on every rule. Every metric and memorandum should lapse in three years unless re-justified with evidence of its effect on the number of transplants. The 2019 removal of outcome requirements for transplant program re-approval was the first reversal of the ratchet in this book (Source Notes S26). It should not have taken twelve years, and it should not be the last. *Test:* sunset clauses written into every new rule.

Group four: protect the front door and show the work (moves 12 and 13)

12. Protect the front door. An independent safety officer in every procurement organization, with the authority to halt a recovery and a direct line to the federal agency; a quarterly public pause log; a written account to every donor family within thirty days. The registry opt-out rate should be tracked weekly and published beside the performance tiers (Source Notes S35; Source Notes S36). *Test:* officers in place and a first public pause log.

13. Show the declines. Each transplant center should publish, annually, its decline reasons and, for kidneys it declined that were later transplanted elsewhere, the outcomes. It is not an accusation. Some declines will turn out to have been right, and the publication will show it. Today a center cannot know whether its caution was wise. *Test:* a first public report from at least one hundred centers.

Group five: the industry and the voice (moves 14 and 15)

14. Open the books on dialysis. A public database of medical-director compensation, physician ownership of vascular access centers and dialysis facilities, and the financial relationships between charities and the firms that fund them; facility-level referral and waitlisting rates displayed beside the star rating; and an independent review by competition authorities of a market in which two firms hold more than three-quarters of the business (Source Notes S46). *Test:* the database and the review.

15. Seat the patients. Every board, advisory committee and trial steering group that makes decisions about allocation, preservation or dialysis should include patients, living donors and donor families with a vote. Not a listening session. A vote. The reason is practical: a body that includes the people who pay for its mistakes makes fewer of them. *Test:* seats filled, and attendance published.

The order

You cannot do fifteen things at once. In my judgment the sequence is this.

First, the cheap and fast. Moves 4, 5, 8, 9, 11, 12 and 13 cost little and can begin by administrative action. They produce information, which is the raw material for everything else.

Second, the structural. Moves 1, 2, 6, 7 and 10 require a rule, a payment or a model, and a year or two. They convert the information into incentives.

Third, the investments. Moves 3, 14 and 15 are slower, and they change who is in the room.

The targets

A plan without a number is a wish. Here are three.

Return the nonuse rate to 18.2 percent within five years. By my arithmetic in Chapter 1, that is about 3,300 more transplants a year, and at the 2021 cost difference, roughly $185 million a year in avoided dialysis cost, as a scenario and not a forecast (Source Notes S2; Source Notes S10).

Cut the median cold time of a kidney by three hours, undoing the added hours of the broader sharing.

Raise the share of kidneys perfused from 39 percent to a figure the network sets in consultation with surgeons and organizations, and publish it, within twelve months.

I have chosen these because they are measurable, because they can be checked by anyone with the registry's tables, and because they are wrong in only one way: if we hit them and nothing else improves, I will have been mistaken, and I will say so.

What must be done

Choose one. It is not possible for any reader to push fifteen levers. It is possible for each of us to choose the one that is closest to where we stand, and to ask the person who controls it, in writing, for a date. The next chapter shows how.

CHAPTER 14

The Ninety Days

> *"While we are postponing, life speeds by."*
>
> Seneca, Letters to Lucilius, letter 1 (Gummere translation, 1917)

In this chapter. *Question:* what can a patient, a care partner, a clinic or a community group do in the next ninety days, without waiting for Congress? *Answer:* ask twelve specific questions, send three specific letters, keep a one-page log, and report what you learn. None of it requires permission. All of it produces the one thing the system lacks, which is information it cannot ignore.

Denise, in my telling, does not give up.

I have kept her waiting for eleven chapters, and she has had a long time to think. In the weeks after the night of the fourteenth offer she read the same material you have, and she did what a fourth-grade teacher does with a list of facts she wants to remember: she turned it into questions. Her ninety days are below. They are invented, as she is, but each step is something a real person can do, and I have tried to keep every one of them within reach of someone with a telephone, a notebook and a Saturday.

Days 1 to 7: the twelve questions

Denise's first appointment after reading was with her transplant coordinator, and she did not ask for a kidney. She asked for information. These are the questions, in the order she asked them. You may copy them.

1. *How many kidney offers has my name received in the last twelve months, and how many did the center accept or decline on my behalf?*

2. *What reasons were recorded for the declines?*

3. *Does this program use offer filters that screen out some kidneys before they reach the surgeon? Which filters apply to me?*

4. *Which categories of kidneys have I agreed to consider: older donors, higher-risk donors, donors with hepatitis C, kidneys from donation after circulatory death? Can I change that?*

5. *Does this center use machine perfusion? What share of the kidneys it accepts are pumped, and who provides the pump?*

6. *How does the program's offer acceptance ratio compare with what is expected for patients like me, and how does its one-year graft survival?* (These are in the public program report from the national registry, and a good coordinator will know where to find them.)

7. *If a higher-risk kidney is offered at night, who decides, and will someone call me?*

8. *Would it help me to be listed at a second center? What would that involve?*

9. *How is my status decided if I am made inactive, and how would I know?*

10. *Is this center taking part in any trials of preservation or perfusion?*

11. *What happened, afterward, to the kidneys this center declined last year: how many were transplanted elsewhere, and how did they do?*

12. *Who represents patients on this program's quality committee?*

Notice what these have in common. None of them demands that the center accept a kidney. None of them accuses anyone. Each asks the program to look at what it already knows. A good center will welcome them, because a good center is proud of its practices. A center that finds them hostile has told you something about itself.

Denise wrote the answers in a notebook, including the ones that were "I'll have to check," and the date she asked. Her coordinator, she noticed, wrote some things down too.

Days 8 to 30: three letters

Letters are underrated. A telephone call can be forgotten; a letter must be answered, and it creates a record.

The first letter went to her transplant center's medical director, politely, with the twelve questions attached. It asked for written answers within thirty days and for a meeting.

The second went to her member of Congress, and it was one page. It said who she was, that she had been on the list for five years, and that she had read that about 29 percent of recovered kidneys were never transplanted. It asked three things: that the representative request from the federal transplant agency a public date for an expedited-placement pathway (move 5), a quarterly offer statement for every candidate (move 8), and a rule on who pays for kidney perfusion (move 1). It closed with the sentence she had learned matters most: *Please reply with a date.*

The third went to the dialysis facility where she spent her Mondays, Wednesdays and Fridays. It asked whether the facility had a policy of referring every new patient for transplant evaluation within ninety days, and what share of its patients had been waitlisted in the last year (move 2). She copied the regional administrator.

If you would rather not write from scratch, here is the form of a model letter. Change the facts to yours.

> Dear [Name]. I am [a patient on the kidney waiting list / a care partner / a living donor] in [town]. I have read that in 2024 about 29 percent of donated kidneys in the United States were never transplanted, up from 18 percent in 2013. I am writing to ask that you request from [agency] a public plan, with dates, to [specific move]. I would be grateful for a written reply by [date, thirty days out]. Sincerely, [Name].

Days 31 to 60: find the others

Denise did not stop at letters. She called the social worker at her dialysis unit and asked whether anyone else had questions like hers. Two did. They became three. They met for an hour on a Saturday at a library and wrote down the answers they had gotten and the ones they had not.

This is the step most people skip, and it is the one that matters most. One patient's questions are an anecdote. Five patients' answers, compared, are a pattern. If three of five centers in a region say they have never been asked about offer filters, that is a finding. If four of five cannot say what share of their kidneys are pumped, that is a finding. The kidney community has been, in one sense, atomized: everyone waits alone, in a different chair.

A group of five can also do what one cannot. It can ask for a meeting with the center's director and expect to get it. It can attend a hospital board's public session. It can write to the local newspaper. And it can find a researcher: many transplant researchers are eager for patients who will read their papers and ask questions.

Days 61 to 90: report

At the end of ninety days Denise sat down with her notebook and made a one-page report. It had four parts.

What I asked, and when. A dated list.

What I was told. Answers, in her own words, with the person's name and role.

What I could not find out. The most important part. The gaps are the evidence.

What I would like next. One request, with a date.

She sent it to her center's director, to her representative, to the head of her dialysis company's regional office, and to the local reporter who covers health. She sent a copy to the patient organization she belongs to. She kept a copy for herself.

A log you can keep

Any patient can keep a simple record. One page, with five columns, is enough.

Date	What I asked	Whom	What I was told	What I'm waiting for

Keep it for ninety days. Bring it to every appointment. The value is not in any single entry. It is in the habit of expecting an answer.

For clinics and community groups

The same plan works from the other side of the table.

A **transplant center** can, this quarter, publish its offer acceptance ratio and one-year outcome in plain language on its own website; list the kinds of kidneys it declines most often and why; and offer every candidate the offer ledger of Chapter 2 before anyone requires it. The first center to do so will be a story.

A **dialysis facility** can report its referral and waitlisting rates to its patients, make a transplant coordinator's name and number visible on the wall, and add a monthly question to its patient survey: *How long did it take you to feel like yourself after your last treatment?*

A **procurement organization** can send every donor family a letter within thirty days of a donation saying what became of each organ, and publish its quarterly pause log.

A **community group**, a church, a union, a veterans' post, a parent association, can host a one-hour session with a local transplant surgeon and a patient, and ask the twelve questions aloud.

What the ninety days are for

They will not change the national system on their own. Denise knows that. She will probably not receive a kidney because of any of this, and I will not pretend that she will.

What the ninety days do is move a piece of the invisible ledger into the visible one. Each question asked, each answer logged, each letter replied to or ignored, is an entry with a name and a date. The system is built to respond to what it can see. For the first time, a few of its customers have started to show it.

Seneca's sentence at the head of the chapter is the reason I wrote this book. We are always postponing something. A patient waiting for a kidney is, in the cruel arithmetic of Chapter 1, waiting on someone else's postponement. It is not asking too much to ask the postponers for a date.

CHAPTER 15

The Last Hour

> *"A partnership not only between those who are living, but between those who are living, those who are dead, and those who are to be born."*
>
> Edmund Burke, Reflections on the Revolution in France (1790)

In this chapter. *Question:* what do we owe a gift, and what would it mean to keep faith with one? *Answer:* a gift is held in trust. A system that wastes it has broken a trust, not merely lost an efficiency. The closing chapter restates the argument in three answers, says what would prove it wrong, and describes the night I want.

There is a moment at the end of every transplant that nobody photographs.

The surgeon has joined the artery and the vein. The clamps come off. For seventeen hours, or nineteen, or thirty, the kidney has lain pale in its bath of cold fluid, and then, over the space of a few seconds, it takes on color, like a face coming back after a faint. A few moments later it begins to make urine. Someone in the room says something, usually something small and practical. The recipient, asleep, is the last to know.

I have never been in that room as the surgeon. I have been near it, and that is enough. What I understand about that moment is that it is the one place in the whole system where the gift becomes whole. Everything this book has described, the numbers, the memo, the scoreboard, the pump, is an argument about how many such moments there will be.

Three answers

I promised at the beginning that you would be able to answer three questions without notes. Here are the answers, as plainly as I can give them.

Where do the kidneys go? About twenty-nine in a hundred of the kidneys recovered in the United States in 2024 were never transplanted. By my arithmetic that is roughly 8,800 kidneys, about twenty-four a day, in a year when 3,743 people died waiting and 4,901 more became too sick to be transplanted (Chapter 1).

Why do they go there? Because the system is built to notice a yes that goes wrong and to ignore a no that goes wrong. A surgeon who accepts a marginal kidney and fails is visible, counted and graded. A surgeon who declines it, and the patient who dies waiting, are neither. Programs under federal scrutiny accepted a fifth fewer offers afterward, including the best kidneys (Chapter 4). A single memorandum cut recoveries within seven weeks (Chapter 5). The clock that damages kidneys has no owner (Chapter 3). The front door has lost public trust (Chapter 6). And the business of dialysis is paid by the chair (Chapter 7).

What would stop it? Fifteen moves, in five groups, each with an owner and a test: pay for what works, make the clock and the list visible, fix the scoreboard, protect the front door and show the work, and open the industry and seat the patients (Chapter 13). Start with the cheap and fast. Choose one. Ask for a date (Chapter 14).

What I may have gotten wrong

A book that argues this hard owes you its own weak points.

I may be overestimating the share of discarded kidneys that were viable. Experts reviewing the 2022 discards estimated 62 percent, and I have used that figure, but it is an estimate and the real figure might be lower. The French comparison is a model. My arithmetic about 8,800 kidneys and $185 million is mine and rests on assumptions I have tried to state.

I may be underestimating the legitimate reasons surgeons decline kidneys. Donors are older and sicker. A delayed graft is a real harm. A transplant program that takes on too many risky kidneys can do damage. The studies I cite on the effect of federal scrutiny are observational, and the memo analyses are early and brief.

I may be wrong about the pump's reach. It is possible that its benefit is concentrated in particular kinds of kidneys, and that the right national figure is lower than I imply. I have said the evidence is strong. I have not said it is complete.

And I cannot say whether the oxygen carriers will work. I hope they do. I have said so, and I have said what a proper trial would look like.

So here is what would prove me wrong. If the nonuse rate falls to 18 percent without any change in the incentives, the scoreboard is not the explanation. If a rigorous trial of the pump fails to replicate, I have misjudged the evidence. If the offer ledger is introduced and nothing changes about the way centers decide, the invisible ledger is not as invisible as I think. I would count each of these as news, and I would rather learn them from you than from a reviewer.

The pig and the ledger

A future is arriving, and Chapter 11 described it: a gene-edited pig kidney that kept a man off dialysis for 271 days, and a cell therapy that let most of a small group of patients stop their anti-rejection drugs. If either becomes routine, it will enter exactly the system this book has described, offered through a computer, accepted or declined by the same programs, graded by the same scoreboard, shipped on the same clock and paid for by the same payers. A technology that arrives in a broken structure does not fix the structure. It inherits it. That is why the fifteen moves matter even if you believe, as I half do, that biotechnology will someday end the shortage.

What we owe a gift

I said at the beginning that I would not talk much about philosophy. I want to break that promise for one page, because I think the philosophy is the point.

A gift is not a transaction. When a family says yes in a corridor at two in the morning, they are not selling. They are giving something that cannot be returned to someone they will never meet. The law reflects this: in the United States it is a federal crime to buy or sell an organ. We have decided, as a country, that the kidney is held outside the market.

But a thing held outside the market is held *in trust*. A trustee has duties. She must not squander what she has been given. She must not hold it for a reason that serves herself. She must account for it. The family, the donor, the recipient and the people still waiting are the beneficiaries, and the system is the trustee. By that standard the discard is not just inefficiency. It is a breach.

The conservative tradition has a name for this duty, and I have found no better. Edmund Burke wrote that a society is a partnership, not only among those who are living but between those who are living, those who are dead, and those who are to be born. A kidney donor is dead. The donor's family is living. The people still on the list are living, and their children, and the people who will one day need a kidney and are not yet born. The partnership is among all of them. Each kidney thrown away is a breach of faith with all of them at once.

The problem is not mysterious. We know what a trustee does. She keeps an honest ledger. She counts what she was given and what she used. She tells the beneficiaries. She does not hide behind a metric. The invisible ledger is a failure of trusteeship, and the remedy is the oldest one there is: make the accounts open.

Francis Bacon, at the beginning of the book, said that time is the greatest innovator. He meant that if you do not change on purpose, change will be done to you. I would add the corollary that this book has been trying to prove. Time is also the greatest discarder. What is not used in time is lost. A kidney has a deadline, and so does every patient on the list, and so does every reader who has come this far and is wondering what to do next.

The night I want

Let me end where I began, with a night.

It is the same Tuesday, and the same phone rings at 11:42 in a transplant center forty minutes from Denise's house. The offer is the same: a fifty-eight-year-old donor, some scarring on a frozen section, nineteen hours of cold time. The surgeon on call is as tired as the first surgeon was, and she is looking at the same screen.

But a few things are different, and none of them is a miracle.

The kidney has been on a pump since recovery, and the numbers from the pump are on the screen beside the biopsy, because her center has a protocol and the procurement organization is paid to provide it. The kidney's journey has been tracked, and the dashboard shows that it was packed forty minutes after recovery and has lost less time than the median. The surgeon knows her center's offer acceptance ratio, and she knows the ratio counts offers for kidneys that are eventually discarded, which makes her more careful about saying no to one she would not say no to again in a year. She knows that her decision will appear, with its reason, in a report that Denise will read.

She says: *Accept.* It is not a certainty. It is a bet, a better-informed one, made in a system that no longer hides the cost of the other bet.

Or she says: *Decline,* and records a reason, and the reason is checked against what becomes of the kidney, and her center learns from it.

Either way, at 7:10 the next morning, Denise's phone shows a message. It says that a kidney was offered for her last night. It says what was decided, and why. And it gives a number to call.

She calls it. She asks what it would take to say yes next time.

That is all I want: a system in which the person at the end of the chain can ask that question, and someone has to answer.

I am not sure we will get there. I am sure that the people who could take us are mostly decent, mostly tired, and mostly waiting for someone to ask. So ask. The clock is running, and it has an owner now. The owner is whoever picks up the phone.

Glossary

Allocation out of sequence (AOOS). Offering an organ to a candidate other than the next one on the computer-generated list, usually after repeated declines.

Cold ischemia time (CIT). The hours an organ spends cooled and without blood flow, between recovery and transplantation.

Delayed graft function (DGF). A transplanted kidney that does not work at once, so that the recipient needs dialysis within the first week.

Donation after circulatory death (DCD). Donation after the heart has stopped, as opposed to after brain death.

Discard / nonuse. A kidney recovered for transplant that is not transplanted into anyone.

Hemodiafiltration (HDF). A form of dialysis that cleans blood by convection as well as diffusion.

Hemoglobin-based oxygen carrier (HBOC, BHOC). A hemoglobin solution, outside red cells, designed to carry oxygen; BHOC denotes a bovine-derived version.

HRSA. Health Resources and Services Administration, the federal agency that oversees the transplant network.

Hypothermic machine perfusion (HMP). A pump that moves cold preservation fluid through a kidney while it waits.

IOTA. Increasing Organ Transplant Access, a mandatory CMS payment model for kidney transplant hospitals that began in July 2025.

KDPI. Kidney Donor Profile Index, a 0 to 100 percentile summarizing the expected quality of a deceased donor kidney; higher means shorter expected life.

Match run. The computer-generated ranked list of candidates for a given organ.

MPSC. The OPTN committee that reviews the performance of transplant programs and organizations.

Offer acceptance ratio (OAR). A program's observed acceptance of kidney offers compared with the number expected for similar candidates.

Offer filters. A tool that lets programs screen out kinds of offers in advance so they are not sent.

OPO. Organ procurement organization: one of 55 regional nonprofits that recover organs and offer them to the list.

OPTN. Organ Procurement and Transplantation Network: writes the rules and runs the matching system.

Orphan innovation. A tool with proven benefit but no party whose budget or metric it fits (the author's term).

Perfusion. The flow of fluid or blood through an organ.

SRTR. Scientific Registry of Transplant Recipients: publishes risk-adjusted program results.

Systems Improvement Agreement (SIA). A corrective plan between CMS and a transplant program whose outcomes fell below expectations.

Waiting list. The national list of candidates registered for a transplant.

Appendix A: Source Notes

Sources cited in the text as “Source Notes S” and a number are described here with the grade I assigned. Grade A is a primary document, registry or peer-reviewed paper whose key figures I could confirm. Grade B is a credible secondary account such as trade press or a news investigation. Grade C is a single source, an advocacy group, a manufacturer or a press release. A dagger (†) marks details only partly confirmed. Where my own arithmetic appears in the text, it is labeled as such.

Note	Grade	What it supports and where it was found	Still to confirm
S1	A†/B	JAMA Network Open (2019), Husain, Mohan and colleagues: about 14 million kidney offers, 2008 to 2015, to more than 350,000 candidates; those who died received a median of 16 offers over 651 days; about 93% of declines attributed to organ or donor quality; about 76% of candidates had at least one viable offer; authors note some declines were correct. Via institutional press release and trade coverage.	Journal article text
S2	A	OPTN/SRTR 2024 Annual Data Report (Overview and Kidney chapters): nonuse of recovered deceased-donor kidneys 29.3% in 2024, 27.9% in 2023, 26.6% in 2022, 18.2% in 2013; biopsied kidneys 40.8% vs 6.4% not biopsied; donors 65 and older 68.9%; record 27,660 kidney transplants in 2024 (27,351 in 2023).	Table numbers
S3	A	Aubert and colleagues, JAMA Internal Medicine 2019;179(10):1365–1374: 2004 to 2014, United States discarded 27,987 of 156,089 recovered kidneys (17.9%), France 2,732 of 29,984 (9.1%); mean donor age of transplanted kidneys about 36.5 (US) and 50.9 (France); under the French model 17,435 (62%) of US discards would have been transplanted; model estimate of 132,445 extra allograft life-years.	Full text

Note	Grade	What it supports and where it was found	Still to confirm
S4	A	Garg and colleagues (MyTEMP), Lancet 2022;400:1693–1703: cluster-randomised Ontario trial of personalised cooler dialysate vs 36.5°C; composite cardiovascular death or hospitalisation 21.4% vs 22.4%, not significant; no difference in intradialytic hypotension; more than 15,000 patients.	Exact patient count, title
S5	A	OPTN/SRTR 2024 Annual Data Report: Kidney: 143,982 adult candidates on the list in 2024.	Table numbers
S6	A	OPTN/SRTR 2024 Annual Data Report: Kidney: record 49,249 adult additions in 2024; 3,743 removals for death and 4,901 for too sick; authors note progress offset by rising nonuse, longer cold ischemia and higher delayed graft function under broader sharing.	Table numbers
S7	B	InvestigateTV (12 February 2026): about one in five donated organs never transplanted; kidney waste higher; Georgia 17% of organs in 2024. Scope of the kidney figure not confirmed.	Full report
S8	C	News report (2026) citing registry counts of 6,419 living kidney donors in 2024 (6,522 in 2025); single source.	OPTN living-donor table
S9	B	Healio (8 August 2023), reporting 2022 OPTN data: 7,540 kidneys (26%) unused; experts estimated 62% viable, implying 4,675 dialysis patients could have received one.	Original analysis
S10	A†	USRDS 2023 Annual Data Report (data through 2021): traditional Medicare cost per person $99,325 (in-center hemodialysis), $86,976 (peritoneal dialysis), $43,913 (functioning transplant).	Table numbers
S11	B	National Kidney Foundation consensus report and a registry analysis of discards: leading recorded reasons “no recipient located” (about 35%) and biopsy findings (about 29%); regulatory and outcome risk cited as drivers of risk aversion.	Primary analysis
S12	B†	Mohan and colleagues, Kidney International (2018): discard rate rose from 14.9% (2000) to 19.0% (2015), by a definition different from the registry nonuse rate.	Authors, exact figures

Note	Grade	What it supports and where it was found	Still to confirm
S13	A†	OPTN-based analysis and Clinical Transplantation 2026 survey introduction: delayed graft function 26% in 2023; 2021 circle-based allocation increased cold ischemia time and delayed graft function.	Table numbers
S14	B	Literature on offer decisions: median seven offers per kidney; 2017 randomized study of 68 physicians (within-center variation about ten times among-center); 2023 survey Fleiss kappa 0.13.	Primary papers
S15	A†	Study of 193 adult kidney programs (PMC12795041): programs accepted a median of 12.5% of non-suboptimal and 7.2% of suboptimal offers; widest variation for cold time over 36 hours; data 2021 to 2023.	Title, authors
S16	A	Reproducibility studies of procurement biopsies: single-center classification match 64% (kappa 0.25; glomerulosclerosis kappa 0.15); repeat biopsies kappa 0.12 to 0.17; second biopsy, not first, associated with graft survival; only glomerulosclerosis over 20% independently associated with discard; 94% frozen sections.	Titles, authors
S17	B	UNOS analysis of provisional acceptances: time spent waiting for a decline can lengthen ischemic time.	Primary document
S18	B	OPTN offer-filter pilot (ATC abstracts and OPTN committee documents): centers avoided 67% of offers; recommended filters could bypass 59%; some centers increased volume with up to 75% fewer offers; HRSA/OPTN board approval of default filters based on acceptance history (2026).	Board resolution text
S19	A†	Debout and colleagues, Kidney International 2015;87(2):343–349 (3,839 recipients, 2000 to 2011): hazard ratio 1.013 per added hour for graft failure, 1.018 for death; 30 hours vs 6 hours about 40% higher risk.	Confirm
S20	A†	Transplant International, June 2026 (doi 10.3389/ti.2026.15840): 74 kidney pairs from older donors; cold ischemia over 12 h vs under 8 h, delayed graft function adjusted odds ratio 6.30 (95% CI 1.52 to 26.06); over 12 h tripled mortality risk; small single center.	Authors

Note	Grade	What it supports and where it was found	Still to confirm
S21	B	Propensity analysis: regional sharing added about 2.9 hours of cold ischemia vs local; OPTN committee statements; network analysis of 106,160 kidneys; median same-hospital cold time 3.3 to 29 hours across 206 centers; weekend procurement more likely discarded.	Primary papers
S22	A/B	HRSA organ-transport working group final report (May 2025, 20 recommendations); FAA organ-transport website (2025); industry briefing (2024) that the contract in force could not require tracking.	Documents
S23	B	Casey Ross, STAT (11 August 2016; republished by PBS NewsHour 13 August 2016): hospitals discarding organs and declining transplants to protect federal performance ratings; Bozorgzadeh study of waitlist removals after 2007 standards; CMS changed benchmarks; 2016 CMS memo; 3,159 kidneys discarded in the year examined, up 20% from 2007.	Original study
S24	A	Federal Register, 30 March 2007: Medicare conditions of participation for approval and re-approval of transplant centers (outcome and clinical-experience requirements based on registry data).	Rule text
S25	A	Bowring and colleagues, American Journal of Transplantation 2018: 14 programs under CMS Systems Improvement Agreements vs 28 matched; acceptance fell 5.9 percentage points (22%); 26.9% to 22.1% vs 33.9% to 44.4%; largest drop for KDPI 0 to 40.	Full text
S26	A/B	Omnibus Burden Reduction final rule, 84 Fed. Reg. 51732 (30 September 2019; effective 29 November 2019) removed re-approval requirements; American Society of Transplant Surgeons commentary (advocacy claim that requirements caused avoidance).	Rule text
S27	A/B	SRTR technical methods and OPTN performance-monitoring documents: offer acceptance ratio used by the MPSC from summer 2023; flag below 0.30 for adults; offers counted only if accepted or declined before an accepted offer (offers for eventually discarded organs excluded).	Current document

Note	Grade	What it supports and where it was found	Still to confirm
S28	A/B	CMS IOTA model fact sheet and 2026 rule: began 1 July 2025; scoring 60 achievement, 20 efficiency (offer acceptance ratio), 20 quality; no first-year results published as of October 2026.	Results when published
S29	A†/B	OPTN/HRSA memorandum of 6 August 2025 on allocation out of sequence; Husain and colleagues (NYU), policy period 6 August to 23 September 2025: lower recovery, transplants, out-of-sequence allocation; about 1,500 fewer transplants a year if sustained; a separate analysis summarized by an advocacy group: recovery fell from about 85 to 76 per day, discard rates did not rise; American Journal of Transplantation 2026 paper on a 30-day elimination of out-of-sequence allocation. Early, short-window analyses.	Peer-reviewed versions
S30	B	Healio (6 April 2026): unilateral out-of-sequence kidney allocations rose 17-fold since 2020.	Original analysis
S31	A†	OPTN-based study (data through 2023): hypothermic machine perfusion used for 39% of deceased-donor kidneys.	Table numbers
S32	A	SRTR 2024 Annual Data Report: nonuse of hepatitis C antibody-positive, NAT-negative kidneys 32.2% in 2024 (49.5% in 2016); Durand and colleagues, Annals of Internal Medicine 2018 (10 recipients) and later trials.	Table numbers
S33	A/B	Blankestijn and colleagues (CONVINCE), NEJM 2023: 1,360 patients, mortality 17.3% vs 21.9%, hazard ratio 0.77 (0.65 to 0.93), median 30 months, mostly fewer infection deaths; Fresenius 5008X FDA 510(k) clearance February 2024 and May 2025; broader US launch 2026; about 160,000 in-center machines (company estimate).	FDA database entry
S34	B	OPTN board meeting, 16 April 2026: update on defining an offer and an expedited kidney placement pathway; no formal action; projects paused pending HRSA review.	Later board minutes
S35	A	Associated Press (28 October 2024): about 170 people a day removed from the national donor registry in the week after coverage of a Kentucky case, ten times the same week of 2023.	Donate Life data

Note	Grade	What it supports and where it was found	Still to confirm
S36	A	HRSA Division of Transplantation report on Kentucky Organ Donor Affiliates (24 March 2025): about 351 cases; 73 where removal should have been reconsidered sooner; 20.8% with features not conducive to donation after circulatory death; at least 28 with no cardiac time of death recorded; HHS summary of concerns in 103 of 351; the organization disputes characterizations.	Final HHS text
S37	A/B	CMS/HHS decertification of Life Alliance Organ Recovery Agency (September 2025), first mid-cycle decertification; reasons cited; organization said it would not appeal.	CMS notice
S38	A/B	OPO tier system (2020 final rule); CMS-3409-P proposed rule (30 January 2026); first recertification cycle ending 2026 on 2024 data. No published roster of Tier 3 organizations found.	Final rule and roster
S39	A†	Transplantation 2026: CMS performance metrics and the disproportionate impact of decertifying OPOs on minority populations.	Authors, details
S40	A/B	Spanish Ministry of Health (January 2026): 51.9 deceased donors per million in 2025, about 6,335 transplants; 2024 record 52.6 per million.	Method of counting
S41	B†	Australian interview study of 49 potential donor families; literature review of donor-family satisfaction.	Titles, authors
S42	A†	Heliyon (2024): analysis of CMS-2728 forms 2007 to 2019 (N = 133,414): 15% not informed about transplant; informed patients listed sooner; chain facilities more likely to inform but not more likely to waitlist; acquired independent facilities less likely to waitlist.	Authors
S43	A	Gander and colleagues, JAMA 2019;322(10):957–973: 1,478,564 patients, 6,511 facilities, 2000 to 2016; for-profit facilities had lower waitlisting; senior author attributes to incentives.	Full text
S44	A†	Kidney Medicine/related 2026 study of pre-dialysis care and access type: more than half of patients started dialysis without access or pre-dialysis nephrology care; unprepared patients more often at for-profit and smaller facilities.	Title, authors

Note	Grade	What it supports and where it was found	Still to confirm
S45	B	CMS ESRD Treatment Choices model evaluations: model in about 31% of regions from 2021; no impact on transplant waitlisting or living-donor transplantation through three years.	Evaluation report
S46	A†	Peer-reviewed market-structure analysis: two largest companies 59.1% (2005) to 77.1% (2019); other estimates about 72% to 80%.	Authors, title
S47	A	MedPAC March 2026 report: aggregate FFS Medicare dialysis margin -0.2% (2023) and 4.5% (2024); recommendation to eliminate the 2027 update; industry letter argues margin overstated by about two points.	Final tables
S48	B/C	DaVita corporate integrity agreement (2014); DOJ settlement over $34 million (July 2024; found via journal reference); 2016 whistleblower allegations about charity premium assistance (disputed); California AB 290 enjoined 2019; claim that about 80% of the charity funding comes from two companies (advocacy).	Court and DOJ records
S49	B	DOJ complaint (2022) against Fresenius Vascular Care alleging 1,288 of 2,303 angioplasties among 60 patients unnecessary (allegation); American Access Care settlement $3,594,791; industry commentary (2026) on fistulas and catheters outside hospitals.	Court records
S50	A†	Registry-based study of home dialysis among incident patients: 6.8% (2010) to 13.3% (2020).	Title, authors
S51	A/B	RaDIANT facility-level trial (Georgia): more than 9,000 patients, 134 facilities; improved referral for evaluation and narrowed racial disparity.	Journal article
S52	A†	American Journal of Transplantation (2021) registry analysis of median graft survival: deceased-donor 8.2 years (1995 to 1999) to about 11.7 years (recent); living-donor 12.1 to about 19.2 years (2014 to 2017).	Authors, titles
S53	C	News report (2026) citing national counts of living kidney donors: 6,290 (2023), 6,419 (2024), 6,522 (2025), 6,867 (2019); single source.	OPTN living-donor table

Note	Grade	What it supports and where it was found	Still to confirm
S54	B	Older analysis of living kidney donation declining after a mid-2000s peak, most among men, Black adults, and younger and lower-income adults; Medical Journal of Australia 2026: 253 living-donor kidney transplants in 2024 vs 354 in 2008.	Primary papers
S55	A/B	Honor Our Living Donors Act (Consolidated Appropriations Act, 2026, P.L. 119-75, enacted 3 February 2026); HRSA notices 2026-13250 (1 July 2026) and 2026-19911; reimbursement up to $6,000 per organ for travel, lost wages, child and elder care; donor-side priority at or below 350% of poverty guidelines; 21% of donors reach the cap (physicians' group letter).	Final notice text, source of 21%
S56	B	National Kidney Registry and American Kidney Fund descriptions of private donor-protection programs; American Kidney Fund report card on state protections (May 2026): gains in several states, little or no progress in many.	Program terms
S57	B/C	End Kidney Deaths Act, H.R. 2687 (119th Congress): refundable credit $10,000 in the donation year and each of four following years for non-directed living kidney donors; kidneys removed after 2026; sunset 2036; about 60 cosponsors (campaign count, March 2026); policy tracker rated chance of passage low.	Congress.gov text
S58	B	National Kidney Registry report of a chain begun in 2008 (30 transplants, 17 hospitals, 11 states, four months); University of Alabama at Birmingham announcements of a chain of 74 donors and 74 recipients, later more than 100 recipients.	Current totals
S59	B	National Kidney Registry figures: 10,000th living-donor transplant; 1,744 transplants in 2024; expected about 30% of US living-donor kidney transplants in 2025. Organization's own figures.	Independent confirmation
S60	B	Live Donor Champion program (Johns Hopkins): pilot of 15 vs 15 matched candidates; 163-candidate follow-up with 81 referrals and about 5.5-fold odds; conference abstracts and registry entry; no randomized trial found.	Published papers

Note	Grade	What it supports and where it was found	Still to confirm
S61	B†	BMC Health Services Research 2023 process-improvement study of living-donor evaluation: about 25% of potential living-donor transplants missed for reasons related to inadequate timing.	Title, authors
S62	A	Moers and colleagues, New England Journal of Medicine 2009;360:7–19: 336 donors, 672 recipients, paired kidneys; delayed graft function 70 vs 89 (adjusted odds ratio 0.57); better one-year graft survival.	Exact survival percentages
S63	A†/C	NEJM correspondence, 5 November 2025 (doi 10.1056/NEJMc2406608): benefit persisted at 10 years; investigators' university release 79% vs 73% functioning; device-maker summary of 27% lower risk of graft failure; 99% follow-up response.	Exact hazard ratio
S64	A	Concepcion and colleagues, Clinical Transplantation 2026 (doi 10.1111/ctr.70637): survey of 88 centers using hypothermic perfusion (35% of centers, 63% of 2024 volume); 74% exclusively OPO-run; OPO refusal the leading barrier (58%); 92% endorse for kidney selection, 90% for reducing delayed graft function; 60% sometimes and 20% would decline a kidney if pumping failed.	Full text
S65	A	Transplant International 2025 (doi 10.3389/ti.2025.15282): Belgium national reimbursement of hypothermic perfusion from September 2022; 242 transplants; delayed graft function 14.4%; donation-after-circulatory-death transplants 90 to 175 per year; estimated savings EUR 3.59 million per year. Retrospective.	Budget model assumptions
S66	A†	Hosgood and colleagues, Nature Medicine 2023: warm perfusion vs cold storage, delayed graft function 60.7% vs 58.5%; American Journal of Transplantation 2026 randomized trial (n=80) adding warm perfusion to cold pump, no improvement; oxygenated pump trial in older donors, no benefit; Nature Reviews Nephrology 2025 review.	Authors, titles
S67	B	Hakai Magazine, Connexion France and company-linked sources on Franck Zal, Roscoff, the lugworm and Hemarina (founded 2007); Zal is a founder and shareholder.	Primary interviews

Note	Grade	What it supports and where it was found	Still to confirm
S68	B/C	Foundation essays on oxygen-carrier history (Autopsy of Five Companies; The Dog Got There First) drawing on reviews and reporting: Baxter, Hemosol, Northfield, Biopure; Oxyglobin approved for dogs in the late 1990s; Biopure bankruptcy 16 July 2009. Secondary; figures such as the 72% mortality increase come from a review and should be checked against original trials.	Original trial reports
S69	A	Natanson and colleagues, JAMA 2008;299(19):2304–2312: 16 trials, 5 products, 3,711 patients; mortality 164 vs 123 (relative risk 1.30, 1.05 to 1.61); myocardial infarction 59 vs 16 (relative risk 2.71, 1.67 to 4.40).	Confirmed in abstract
S70	B/C	OXYOP study of HEMO2life (60 paired kidneys, French centers, 2016 to 2018); 4-year patient survival 98.3% vs 86% (p=0.016, company-linked source); control kidneys had longer cold time by design; Le Meur and colleagues, Artificial Organs 2022 (conflict of interest disclosed); OxyOp2 randomized trial protocol, Trials 2023; sources differ on enrollment (60 vs 120).	Journal text, results of OxyOp2
S71	C	Cold Ischemia Foundation pages on BHOC Therapeutics and BHOC Transplant: described as a research concept; sponsors state BHOC has not been shown to change transplant outcomes.	Sponsor documents
S72	A/B	Massachusetts General Hospital and Harvard Medical School accounts of Tim Andrews (pig kidney January 2025; 271 days; removed October 2025; human kidney January 2026); Nature Medicine 2025 immune profiling report; American Journal of Transplantation 2026 review.	Peer-reviewed case report
S73	C†	News coverage of Richard Slayman (Massachusetts General Hospital, March 2024; died May 2024; hospital reported no indication the transplant caused death) and Towana Looney (NYU Langone, November 2024; kidney removed spring 2025 for rejection). Not reconfirmed from primary sources.	Hospital releases
S74	A/B	FDA clearance of living-patient pig-kidney trials (autumn 2025); eGenesis trial approval (September 2025); United Therapeutics EXPAND trial (NCT06878560; ten gene edits; first transplant announced 3 November 2025; 24-week monitoring).	Trial registry

Note	Grade	What it supports and where it was found	Still to confirm
S75	A/B	Medeor MDR-101 phase 3 (American Journal of Transplantation, July 2025; interim 2023): 20 treated, 10 control; 16 of 19 off immunosuppression at two years (interim); about 95% discontinued all immunosuppression about one year; 75% off more than two years; four resumed.	Journal article
S76	B	HCPLive (23 April 2026) interview with a transplant nephrologist on tolerance as the frontier of kidney care.	Primary interview
S77	B	Atul Gawande, "Slow Ideas," The New Yorker (29 July 2013), via summaries; histories of anesthesia (1846) and of Semmelweis (1847), widely retold; figures commonly cited and not independently confirmed here.	Primary texts

What is thin or unfinished

Several matters deserve a plain statement. The count of unused kidneys in 2024 is my arithmetic from registry rates and should be replaced by the registry's table. The analyses of the August 2025 memorandum are early, brief and in part summarized by trade or advocacy sources. The ten-year follow-up of the machine-perfusion trial reached me through the investigators' university and the device maker. The oxygen-carrier evidence is early, small and partly company-sourced. The allegations about dialysis companies and vascular access centers are allegations; I did not find final judgments. The outcome of the April 2026 board discussion of out-of-sequence allocation, and the 2026 OPO recertification results, had not been published when this book was written.

Appendix B: References

Entries marked † have bibliographic details that are partly confirmed. Sources described only in the Source Notes are not repeated here.

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